

PM/APM/HPM Service - 2

Process Manager Test System

PM13-610

07/00

PM/APM/HPM Service - 2

Process Manager Test System

Publication No. PM13-610

Release 610
Date 07/00

Notices and Trademarks

Copyright 1999, 2000 by Honeywell Inc.
Release 610 July, 2000

While this information is presented in good faith and believed to be accurate, Honeywell disclaims the implied warranties of merchantability and fitness for a particular purpose and makes no express warranties except as may be stated in its written agreement with and for its customers.

In no event is Honeywell liable to anyone for any indirect, special or consequential damages. The information and specifications in this document are subject to change without notice.

Honeywell, and **TotalPlant** are U.S. registered trademarks.

Other brand or product names are trademarks of their respective owners.

Honeywell Inc.
Industrial Automation and Control
16404 North Black Canyon Highway
Phoenix, AZ 85053
1-800 852-3211

About This Publication

This publication provides instructions for use by the High-Performance Process Manager maintenance personnel. It will help you to maintain, test, troubleshoot, and repair the equipment.

This publication supports TotalPlant Solution (TPS) System network software Release 600. TPS is the evolution of TDC 3000^X. The Universal Station (US) displays shown in this manual are produced by R520 software.

This publication supports CE Compliant equipment. Equipment designated as “CE Compliant” complies with the European Union EMC and its health and safety directives. Equipment entering European countries after January 1, 1996 require this type of compliance which is identified by the “CE Mark” symbol.

A change bar is used to indicate that a paragraph, table, or illustration has been changed or added because of updated information. A change bar is not used to indicate a correction because of a minor typographical error.

Technical Assistance

If you need assistance

If you need technical assistance, contact your local Honeywell Service Organization, as explained in the following paragraphs.

International customers

Outside of the United States, contact your local Honeywell Service Organization. If you are not sure of the location or telephone number, call your Honeywell representative for information.

Customers inside the United States

Within the United States, call the Technical Assistance Center (TAC) at the toll free number 1-800-822-7673.

Arizona customers

Within Arizona, the local number for TAC is 602-313-5558.

Services provided

Calls to TAC are answered by a dispatcher from 7:00 A.M. to 5:00 P.M., Mountain Standard Time (6:00 A.M. to 4:00 P.M. when daylight saving time is in effect). Outside of these hours, emergency calls—those which affect your ability to control or view a process—will be received by an answering service, and returned within one hour. TAC maintains its own TPS system, and frequently can duplicate equipment problems.

Time saving tip

It is a good idea to make specific notes about the problem before making the call. This will help to reduce delays and expedite answers.

Standard Symbols

Scope

The standard symbols used in this publication are defined as follows.

ATTENTION

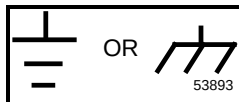
Notes inform the reader about information that is required, but not immediately evident.

CAUTION

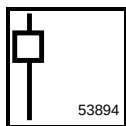
Cautions tell the user that damage may occur to equipment if proper care is not exercised.

WARNING

Warnings tell the reader that potential personal harm or serious economic loss may happen if instructions are not followed.



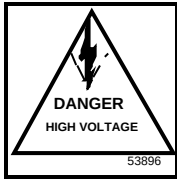
Ground connection to building safety ground.



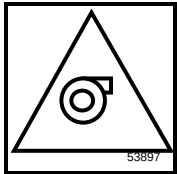
Ground stake for building safety ground.



Electrical Shock Hazard—can be lethal.



Electrical Shock Hazard—can be lethal.



Rotating Fan—can cause personal injury.



Caution—refer to the appropriate installation document.

References

Publication Title	Binder Title	Binder Number
High-Performance Process Manager Specification and Technical Data	System Summary - 2	TPS 3010-2
High-Performance Process Manager Installation	Implementation/ High-Performance Process Manager - 3	TPS 3066-3
High-Performance Process Manager Checkout	Implementation/ High-Performance Process Manager - 3	TPS 3066-3
High-Performance Process Manager Service	PM/APM/HPM Service - 1	TPS 3061-1
Process Manager I/O Specification and Technical Data	System Summary - 2	TPS 3010-2
Process Manager I/O Installation	Implementation/ High-Performance Process Manager - 3	TPS 3066-3
TPS System Site Planning	System Site Planning - 1	TPS 3020-1
Universal Control Network Specification and Technical Data	System Summary - 2	TPS 3010-2
Universal Control Network Planning	System Site Planning - 1	TPS 3020-1
Universal Control Network Installation	Installation/Universal Control Network	TPS 3041
Universal Control Network Guidelines	Installation/Universal Control Network	TPS 3041
Local Control Network Planning	System Site Planning - 1	TPS 3020-1
LCN System Installation	LCN Installation	TPS 3025
LCN System Checkout	LCN Installation	TPS 3025
LCN Guidelines - Implementation, Troubleshooting, and Service	LCN Installation	TPS 3025

Table of Contents

TABLE OF CONTENTS.....	VII
SECTION 1 - INTRODUCTION.....	13
1.1 WHAT PMTS IS AND HOW IT WORKS	13
1.1.1 PMTS Packaging and Loading.....	13
1.1.2 Test Program Execution.....	15
1.1.3 I/O Link Addresses	15
1.1.4 Configuration Program	17
1.1.5 Selecting IOPs for Testing.....	17
1.1.6 Alarm Codes Conventions.....	17
1.1.7 Selecting the I/O Link Cable for Testing	18
1.1.8 Time Synchronization Messages.....	18
1.2 USES OF PMTS.....	18
1.3 REFERENCES.....	19
SECTION 2 - ANALOG OUTPUT MODULE TESTS (AOMT).....	21
2.1 AOMT TEST ELEMENTS	21
2.2 USE INFORMATION.....	21
2.3 SETUP PARAMETERS.....	21
2.3.1 AOMT-Specific Parameters.....	21
2.3.2 General Parameters	23
2.4 PRESET PARAMETER VALUES FOR AOMT	24
2.5 MODE CHARACTERISTICS FOR AOMT	25
2.6 INDIVIDUAL AOMT TESTS.....	25
SECTION 3 - DIGITAL INPUT MODULE TESTS (DIMT).....	29
3.1 DIMT TEST ELEMENTS	29
3.2 USE INFORMATION.....	29
3.3 SETUP PARAMETERS.....	29
3.3.1 DIMT-Specific Parameters	29
3.3.2 General Parameters	30
3.4 PRESET PARAMETER VALUES FOR DIMT	30
3.5 MODE CHARACTERISTICS FOR DIMT	31

3.6 INDIVIDUAL DIMT TESTS.....	31
SECTION 4 - DIGITAL OUTPUT MODULE TESTS (DOMT).....	33
4.1 DOMT TEST ELEMENTS	33
4.2 USE INFORMATION.....	33
4.3 SETUP PARAMETERS.....	33
4.3.1 DOMT-Specific Parameters	33
4.3.2 General Parameters	35
4.4 PRESET PARAMETER VALUES FOR DOMT	35
4.5 MODE CHARACTERISTICS FOR DOMT	36
4.6 INDIVIDUAL DOMT TESTS	36
SECTION 5 - HIGH LEVEL ANALOG INPUT MODULE TESTS (HLAI)	37
5.1 HLAI TEST ELEMENTS.....	37
5.2 USE INFORMATION.....	37
5.3 SETUP PARAMETERS.....	37
5.3.1 HLAI-Specific Parameters	37
5.3.2 General Parameters	39
5.4 PRESET PARAMETER VALUES FOR HLAI	40
5.5 MODE CHARACTERISTICS FOR HLAI.....	41
5.6 INDIVIDUAL HLAI TESTS.....	41
SECTION 6 - LOW LEVEL ANALOG INPUT MODULE TESTS (LLAI)	43
6.1 LLAi TEST ELEMENTS	43
6.2 USE INFORMATION.....	43
6.3 SETUP PARAMETERS.....	43
6.3.1 LLAi-Specific Parameters.....	43
6.3.2 General Parameters	46
6.4 PRESET PARAMETER VALUES FOR LLAi	47
6.5 MODE CHARACTERISTICS FOR LLAi.....	48
6.6 INDIVIDUAL LLAi TESTS	49
SECTION 7 - SMART TRANSMITTER INTERFACE MODULE TESTS (STIM).....	51
7.1 STIM TEST ELEMENTS	51
7.2 USE INFORMATION.....	51

7.3 SETUP PARAMETERS.....	52
7.3.1 STIM-Specific Parameters	52
7.3.2 General Parameters	55
7.4 PRESET PARAMETER VALUES FOR STIM	56
7.5 MODE CHARACTERISTICS FOR STIM	56
7.6 INDIVIDUAL STIM TESTS	57
SECTION 8 - PULSE INPUT MODULE TESTS (PIMT)	59
8.1 PIMT TEST ELEMENTS	59
8.2 USE INFORMATION.....	59
8.3 SETUP PARAMETERS.....	60
8.3.1 PIMT-Specific Parameters	60
8.3.2 General Parameters	61
8.4 PRESET PARAMETER VALUES FOR PIMT	61
8.5 MODE CHARACTERISTICS FOR PIMT	62
8.6 INDIVIDUAL PIMT TESTS	63
8.7 PIMT REPORTS AND ALARMS	64
8.7.1 Total Count Report	64
8.7.2 Pulse Rate Report	65
8.7.3 Point No Match Report	65
8.7.4 Point Out of Tolerance Alarm	66
8.7.5 Point Return to Within Tolerance Report.....	66
8.7.6 Out of Tolerance Report.....	67
SECTION 9 - LOW LEVEL MUX ANALOG INPUT MODULE TESTS (LLMX)	69
9.1 LLMX TEST ELEMENTS	69
9.2 USE INFORMATION.....	69
9.3 SETUP PARAMETERS.....	70
9.3.1 LLMX-Specific Parameters.....	70
9.3.2 General Parameters	73
9.4 PRESET PARAMETER VALUES FOR LLMX	73
9.5 MODE CHARACTERISTICS FOR LLMX.....	74
9.6 INDIVIDUAL LLMX TESTS	75
SECTION 10 - INPUT/OUTPUT PROCESSOR MONITOR (IOPM)	77

10.1 IOPM TEST ELEMENTS.....	77
10.2 USE INFORMATION.....	77
10.3 SETUP PARAMETERS.....	77
10.3.1 IOPM-Specific Parameters—none	77
10.3.2 General Parameters	77
10.4 PRESET PARAMETER VALUES FOR IOPM.....	78
10.5 MODE CHARACTERISTICS FOR IOPM.....	78
10.6 INDIVIDUAL IOPM TESTS	78
SECTION 11 - ANALOG INPUT HISTOGRAM PROGRAM (HIST)	79
11.1 HIST TEST ELEMENTS.....	79
11.2 USE INFORMATION.....	79
11.3 SETUP PARAMETERS.....	79
11.3.1 HIST-Specific Parameters	79
11.3.2 General Parameters	82
11.4 PRESET PARAMETER VALUES FOR HIST.....	83
11.5 MODE CHARACTERISTICS FOR HIST	84
11.6 INDIVIDUAL HIST TESTS	84
11.7 HIST HISTOGRAM REPORTS.....	85
11.7.1 Histogram Summary Report.....	85
11.7.2 Histogram Tabular Report.....	86
SECTION 12 - DIGITAL INPUT SEQUENCE OF EVENTS (DSOE).....	87
12.1 DSOE TEST ELEMENTS.....	87
12.2 USE INFORMATION.....	87
12.3 SETUP PARAMETERS.....	87
12.3.1 DISOE Specific Parameters	87
12.3.2 General Parameters	89
12.4 PRESET PARAMETER VALUES FOR DSOE.....	90
12.5 MODE CHARACTERISTICS FOR DSOE.....	90
12.6 INDIVIDUAL DSOE TESTS	91
12.7 EVENT CIRCULAR LIST REPORT	94
12.8 DSOE MESSAGES	94
12.9 COMMUNICATION ERRORS.....	95

12.10 OVERFLOW/EVENTS LOST MESSAGE	95
SECTION 13 - SERIAL INTERFACE MODULE TESTS (SIMT)	97
13.1 SIMT TEST ELEMENTS	97
13.2 USE INFORMATION.....	97
13.3 SETUP PARAMETERS.....	97
13.3.1 SIMT-Specific Parameters	97
13.3.2 General Parameters	98
13.4 PRESET PARAMETER VALUES FOR SIMT	98
13.5 MODE CHARACTERISTICS FOR SIMT	99
13.6 INDIVIDUAL SIMT TESTS	99
13.7 SIMT MESSAGES.....	102
13.8 UDC 6000 SETUP.....	104
13.8.1 Operational Description.....	105
SECTION 14 - FIELDBUS TESTS (FBUS).....	109
14.1 FBUS TEST ELEMENTS	109
14.2 USE INFORMATION.....	109
14.3 SETUP PARAMETERS.....	109
14.3.1 FBUS-Specific Parameters	109
14.3.2 General Parameters	110
14.4 PRESET PARAMETER VALUES FOR FBUS	110
14.5 MODE CHARACTERISTICS FOR FBUS	111
14.6 INDIVIDUAL FBUS TESTS	111
APPENDIX A - HARDWARE INTERACTIONS AND ALARM CODES	113
A.1 COMMON EVENTS AND ALARM CODE CONVENTIONS.....	113
A.1.1 Hardware Interactions of Common Events	113
A.1.2 Alarm Codes Conventions	114
A.2 AOMT INTERACTIONS AND ALARM CODES.....	115
A.2.1 Hardware Interactions of AOMT	115
A.2.2 AOMT Alarm Codes.....	116
A.3 DIMT INTERACTIONS AND ALARM CODES.....	118
A.3.1 Hardware Interactions of DIMT	118
A.3.2 DIMT Alarm Codes	119

A.4 DOMT INTERACTIONS AND ALARM CODES.....	120
A.4.1 Hardware Interactions of DOMT	120
A.4.2 DOMT Alarm Codes.....	122
A.5 HLAI INTERACTIONS AND ALARM CODES	123
A.5.1 Hardware Interactions of HLAI.....	123
A.5.2 HLAI Alarm Codes	123
A.6 LLAI INTERACTIONS AND ALARM CODES.....	124
A.6.1 Hardware Interactions of LLAI	124
A.6.2 LLAI Alarm Codes.....	125
A.7 STIM INTERACTIONS AND ALARM CODES.....	126
A.7.1 Hardware Interactions of STIM	126
A.7.2 STIM Alarm Codes.....	127
A.8 PIMT INTERACTIONS AND ALARM CODES.....	128
A.8.1 Hardware Interactions of PIMT	128
A.8.2 PIMT Alarm Codes.....	129
A.9 LLMX INTERACTIONS AND ALARM CODES.....	130
A.9.1 Hardware Interactions of LLMX	130
A.9.2 LLMX Alarm Codes.....	131
A.10 IOPM INTERACTIONS AND ALARM CODES	131
A.10.1 Hardware Interactions of IOPM.....	131
A.10.2 IOPM Alarm Codes	131
A.11 HIST INTERACTIONS AND ALARM CODES	132
A.11.1 Hardware Interactions of HIST.....	132
A.11.2 HIST Alarm Codes	132
A.12 DSOE INTERACTIONS AND ALARM CODES	133
A.12.1 HARDWARE INTERACTIONS OF DSOE.....	133
A.12.2 DSOE ALARM CODES.....	134
A.13 SIMT ALARMS.....	134

Section 1 - INTRODUCTION

This section introduces you to the characteristics and uses of PMTS.

1.1 WHAT PMTS IS AND HOW IT WORKS

The Process Manager Test System (PMTS) is a collection of test programs that work under the Process Manager Test Executive (PMEX) and are used to verify the correct operation of individual I/O Processor (IOP) cards connected to **TotalPlant** Solution (TPS) System Process Manager Modules (PMMs), Advanced Process Manager Modules (APMMs), and High-Performance Process Manager Modules (HPMMs). Each test program is concerned with a specific I/O Processor card type.

Table 1-1 lists the PMTS test programs (called HVTPs) by the 4-character names accepted by PMEX.

Table 1-1 — PMTS Test Program Names

AOMT	—	Analog Output Module Tests
DIMT	—	Digital Input Module Tests
DOMT	—	Digital Output Module Tests
HIST	—	Analog Input Histogram Program
HLAI	—	High Level Analog Input Module Tests
IOPM	—	Input/Output Processor Monitor
LLAI	—	Low Level Analog Input Module Tests
LLMX	—	Low Level Mux Analog Input Module Tests
PIMT	—	Pulse Input Module Tests
SIMT	—	Serial Interface Module Tests
STIM	—	Smart Transmitter Interface Tests
DSOE	—	Digital Input Sequence Of Events Tests
FBUS	—	Fieldbus Tests

The test programs are subdivided into individual tests that are designed to exercise specific functions within the IOP. You control the operation of PMTS by keyboard entry of commands that specify the IOP(s) to be tested and the number and sequence of tests to be run.

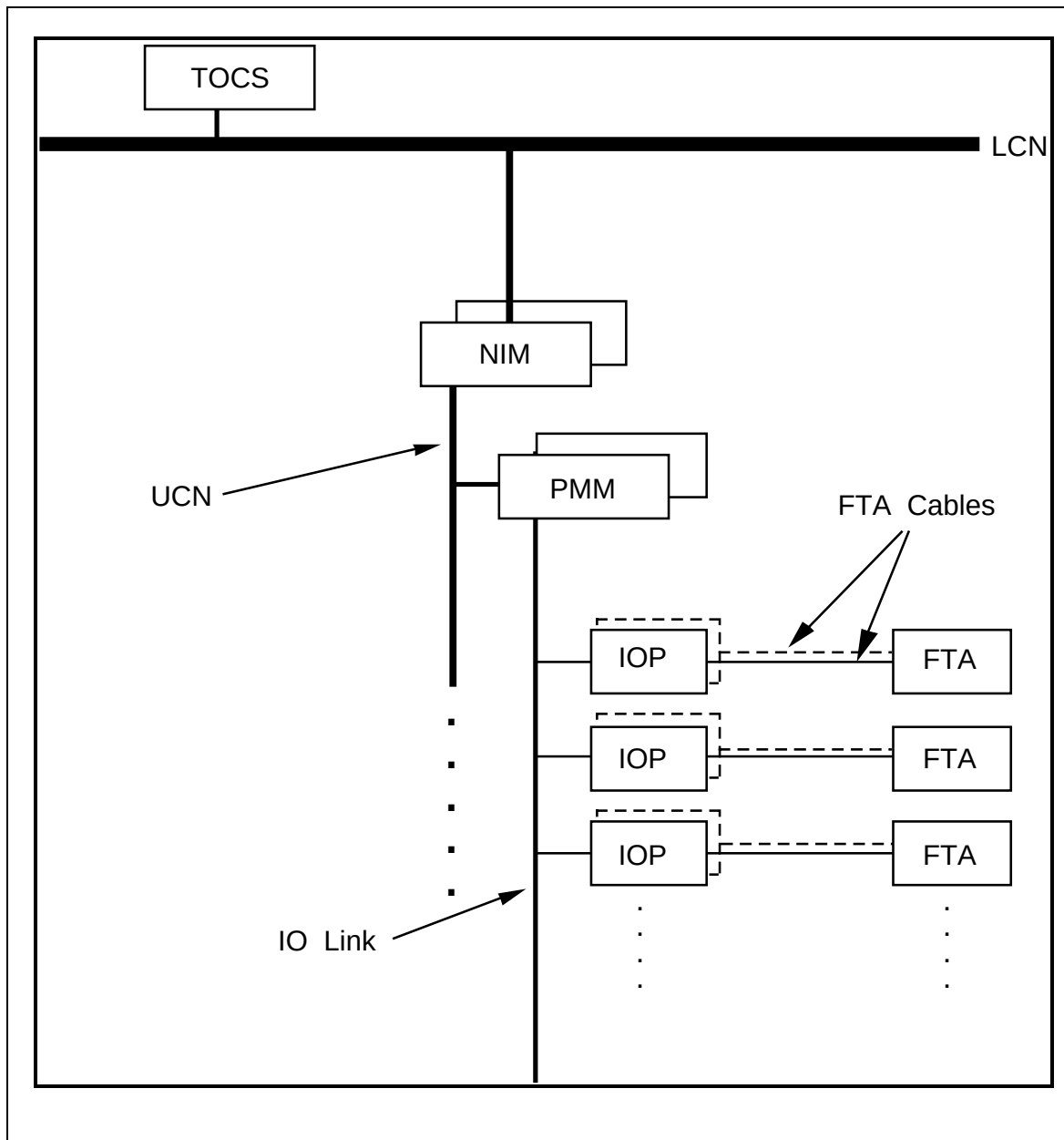
1.1.1 PMTS Packaging and Loading

Each PMTS test program documented in this manual is packaged in one of four load images (named PMTS, PMT2, PMT3, and HIST) that also include the Process Manager Test Executive (PMEX). The PMTS load image is on floppy diskette 51150725, the PMT2 load image is on floppy diskette 51150726, PMT3 is on floppy diskette 51150727, and the HIST load image is on floppy diskette 51150743. All four load images are on test systems cartridge disk 51152049. Follow the loading and test system setup instructions found in the *Process Manager Test Executive* manual.

The PMTS load image contains test programs AOMT, HLAI, IOPM, LLAI, and LLMX.

The PMT2 load image contains test programs DIMIT, DOMT, IOPM, STIM, and PIMT. The PMT3 load image contains the test programs DSOE, IOPM and SIMT. The HIST load image contains the test program HIST.

Figure 1-1 — Process Manager Test System Overview



1.1.2 Test Program Execution

Unlike HVTS and MMTS, PMTS does not run more than one numbered test within a test program at a time (SIMT is the only exception). It allows only one test at a time in the test list for each HVTP. More than one test can be specified in the test list when in Exerciser Mode, but only the single exerciser test will be run. Multiple HVTPs (with one test each) can run at the same time in Module or Exerciser mode. See "Mode Characteristics" in the specific test program chapters for more information.

CAUTION

When testing a redundant Advanced Process Manager, do not run either PMTS, PMT2, or PMT3 tests on both APMMs at the same time. Two APMMs commanding the same IOP produces unpredictable test results.

1.1.3 I/O Link Addresses

The PMTS notation for I/O Link addresses is referred to as "FSS." This names the IOP board file and slot address, where the file (F) value can range from 1-8 and the slot (SS) value can range from 01-15. For example, the last slot of the first file is 115 and the first slot of the second file is 201.

The allocation of slots within files between APMM cards (Modem, Comm, Control, etc.) and IOP cards is controlled by whether or not the APM is redundant (or is configured for future redundancy). The allocation of slots reserved for APMM cards and the slots available for IOP cards varies as follows:

APM Type	File 1		File 2		File 3	
	APMM Slots	IOP Slots	APMM Slots	IOP Slots	APMM Slots	IOP Slots
Nonredundant	1-5	6-15	—	1-15	—	1-15
Left-right Redundant	1-10	11-15	—	1-15	—	1-15
Up-down Redundant	1-5	6-15	1-5	6-15	—	1-15

Thus, assuming a APM to be in files 1-3, the legitimate IOP address ranges are:

nonredundant APM—106-115, 201-215, and 301-315 (40 IOPs maximum)

left-right redundant APM—111-115, 201-215, and 301-315 (35 IOPs maximum)

up-down redundant APM—106-115, 206-215, and 301-315 (35 IOPs maximum)

	File 1		File 2		File 3	
HPM Type (15 slot chassis)	HPMM Slots	IOP Slots	HPMM Slots	IOP Slots	HPMM Slots	IOP Slots
Nonredundant	1-2	3-15	—	1-15	—	1-15
Left-right Redundant	n/a	n/a	n/a	n/a	n/a	n/a
Up-down Redundant	1-5	6-15	1-5	6-15	—	1-15

	File 1		File 2		File 3	
HPM Type (7 slot chassis)	HPMM Slots	IOP Slots	HPMM Slots	IOP Slots	HPMM Slots	IOP Slots
Nonredundant	1-2	3-7	—	11-15	—	1-7
Left-right Redundant	1-2	3-7	9-10	11-15	—	1-7
Up-down Redundant	n/a	n/a	n/a	n/a	n/a	n/a

Thus, assuming an HPM to be in files 1-3, the legitimate IOP address ranges are:

(15-slot chassis)

nonredundant HPM—103-115, 201-215, and 301-315 (43 IOPs maximum)

left-right redundant HPM— Not Applicable for HPM

up-down redundant HPM—103-115, 203-215, and 301-315 (41 IOPs maximum)

(7-slot chassis)

nonredundant HPM—103-107, 201-215, and 301-315 (35 IOPs maximum)

left-right redundant HPM—103-107, 111-115, 201-215, and 301-315 (40 IOPs maximum)

up-down redundant HPM— not Applicable for HPM

1.1.4 Configuration Program

The configuration program determines which addresses on the I/O Link respond, then determines the hardware type, revision level, and status of each responder and places this information in a configuration table for use by the TOCS.

The configuration program runs automatically at system startup, and thereafter on the command `AUTO_CONFIGURATION` (which can be abbreviated as `AUT`). See the PMEX manual for details.

To display the configuration table, use the `CONFIGURATION (CON)` command.

1.1.5 Selecting IOPs for Testing

Once the configuration program has run you can add or delete addresses to/from the configuration table by use of the commands `ADD_IOP` and `DEL_IOP`. IOPs can be listed individually in these commands and you can use "dash notation" for ranges of addresses. For example, the command line `DEL_IOP 110, 201-305` would disable testing on IOP 110 as well as all IOPs in file 2 and IOPs between slots 1 and 5 of file 3.

Pairs of IOPs of the same type are selected for redundancy testing by using the `PARTNER` command. For example, `PAR 115, 116` assigns like IOPs existing in file 1, slots 15 and 16 as a redundant pair.

1.1.6 Alarm Codes Conventions

In PMTS tests, the alarm text includes, among other things, what the test program was doing during the failure, what went wrong, and pertinent status information. Alarm codes included with the text usually are not as helpful in diagnosing a hardware failure as is the text of the message.

See Appendix A for additional information on Alarm codes and hardware interactions.

1.1.7 Selecting the I/O Link Cable for Testing

A new command, LINK_IO, has been added to the Process Manager Test Executive (PMEX) to support I/O Link cable selection during PMTS testing. LINK_IO (abbreviated as LIN) can be set to the values A, B, or SWITCH (S) with “A” being the default setting. LIN SWITCH (or S) causes the I/O Link cable to switch once a minute.

The actual cable selection is performed by a PMTS task, IOLM (the IO Link Manager), which is hidden from the operator. All IOPs in the IOP Configuration Table are commanded to “listen” to the selected cable (transmissions are always sent to both cables). All IOPs in the IOP Configuration Table are scanned following each change in the cable selection or once a minute when cable selection is fixed on A or B to ensure that the IOPs are “listening” to the selected cable. If an IOP is found to be listening to the “wrong” cable, IOLM will report it in an alarm. With redundant PMMs/APMMs, cable selection and scanning are done with only one PMM/APMM or the other, depending upon which one completed the “load” operation first (this is usually not apparent to the operator). Resetting IOPs or the redundant partner PMM/APMM can cause an alarm.

1.1.8 Time Synchronization Messages

A Warning may occur when the user loads a PM when LSOE is running on the NIM, and any test program is loaded (PMTS, PMT2, or PMT3) on an APMM/HPMM. This warning occurs whenever more than one NIM is sending Time Synchronization, or when a newly loaded APM/HPM begins to send Time Synchronization unexpectedly.

MESSAGE

```
OPERATOR ERROR - TOO MANY NIMs ARE SENDING TIME STAMPS ON THIS
UCN!!! ABORT "LSOE" IN BOTH NIMs ON THIS UCN, WAIT 30 SECONDS
THEN RESTART ONE OF THEM!
```

This is to be treated as a user warning. LSOE is to be halted at the TOCs for a period of 30 seconds and testing is to resume with LSOE being reinitialized.

1.2 USES OF PMTS

PMTS is useful during both system installation and initial hardware validation and, in a limited sense, for troubleshooting during on-line system operation. System validation is accomplished before loading operational software and, in that instance, PMTS has unconstrained use of the system hardware. If PMTS is to be used for troubleshooting during normal operation, the selected TOCS and the NIM (and therefore the PMMs and IOPs to be tested) must be removed from service before PMTS can be loaded.

1.3 REFERENCES

The following are other TPS 3000 test program publications that will be of use during installation and maintenance of TPS 3000 Process Manager Modules.

Publication Title	Publication Number	Binder Title	Binder Number
Test System Executive	SW13-510	LCN Service - 3	TPS 3060-3
Process Manager Test Executive (PMEX)	PM13-620	PM/APM/HPM Service - 2	TPS 3061-2
Process Manager Module Test System (PMMTS)	PM13-505	PM/APM/HPM Service - 2	TPS 3061-2

Section 2 - ANALOG OUTPUT MODULE TESTS (AOMT)

2.1 AOMT TEST ELEMENTS

This program tests Analog Output Modules and their associated FTAs that are connected to a selected PMM, APMM, HPMM.

2.2 USE INFORMATION

The tests in this program run on a single analog output IOP, if a value for IOP_ADDRESS is supplied, or on all configured AO IOPs, if that parameter value equals zero.

A pass for any of these tests consists of writing a value or a series of values to all specified points (card slots) on all specified available AO cards. The values are random or operator-specified.

The testing of redundant IOPs is controlled by the parameter FTA_CABLE_SWITCH. When this parameter is set to AUTO, switching is automatic. When it is set to A_CABLE or B_CABLE, only that IOP is tested.

2.3 SETUP PARAMETERS

2.3.1 AOMT-Specific Parameters

The following operator-selected values are used by one or more of the tests in this program.

IOP_ADDRESS	File and Slot address of an analog output IOP to be tested (the IOP must appear in the configuration table). See headings 1.1.3 through 1.1.5 for information on IOP addressing and configuration. A value of 000 for this parameter calls for the testing of all AO IOPs contained in the configuration table.
POINT_SLOT_MASK	Points/slots on an AO IOP that are to be tested. There are 8 or 16 slots on an AO IOP and each slot to be tested must be specified. For example, the entry 1, 3-6, 0 indicates that slots 1,3,4,5, and 6 are to be tested. The terminal 0 indicates the end of test slot input and must be included.
UNITS_OF MEASURE	Name (1 to 20 characters) of the units of measure associated with the operator-specified test input or output values for the AO IOP device. Example unit of measure names are voltage, current, and percent.
REF_100_PERCENT_VALUE	A floating point value that specifies the output in terms of the Units of Measure when the AO IOP output value is at its 100% point.
REF_0_PERCENT_VALUE	A floating point value that specifies the output in terms of the Units of Measure when the AO IOP output value is at its 0% point.
STARTING_VALUE	A floating point value that specifies the initial test value of

	the AO IOP output in terms of the Units of Measure.
MAXIMUM_VALUE	A floating point value that specifies the maximum test value of the AO IOP output in terms of the Units of Measure.
MINIMUM_VALUE	A floating point value that specifies the minimum test value of the AO IOP output in terms of the Units of Measure.
STEP_VALUE	A floating point value that specifies the incremental change of test value of the AO IOP output in terms of the Units of Measure.
STEP_TIME	An integer value (1-20) that specifies the length of time (in seconds) that the test value of the AO IOP output is to be held at its current value before it is changed.
MAX_REACHED_ACTION	Specifies the action to be taken when the AO IOP output reaches the specified Maximum Value. Allowed values are NO_ACTION REVERSE_STEP_SIGN GO_TO_MAX GO_TO_MIN GO_TO_OTHER_MAX_OR_MIN GO_TO_START_VALUE SINE_WAVE
MIN_REACHED_ACTION	Specifies the action to be taken when the AO IOP output reaches the specified Minimum Value. Allowed values are NO_ACTION REVERSE_STEP_SIGN GO_TO_MAX GO_TO_MIN GO_TO_OTHER_MAX_OR_MIN GO_TO_START_VALUE SINE_WAVE
START_REACHED_ACTION	Specifies the action to be taken when the AO IOP output reaches the specified Starting Value. Allowed values are NO_ACTION REVERSE_STEP_SIGN GO_TO_MAX GO_TO_MIN

	GO_TO_OTHER_MAX_OR_MIN
	GO_TO_START_VALUE
	SINE_WAVE
ENABLE or DISABLE	Allows the operator to enable or disable specific functions associated with execution of tests. The only allowed entry value is DEMAND—demand is enabled by the operator after he has changed STARTING_VALUE. This triggers a new output by Test 3; the program then disables demand.
PATTERN	Specifies whether test 2 is to use a random or "wig-wag" (29.64% FS/84.30% FS) pattern. Allowed values are RANDOM WIG_WAG
FTA_CABLE_SWITCH	Applies to redundant IOP pairs only. Allows the test operator—by specifying the FTA cable position—to select which IOP of a redundant pair will be tested. Selection of AUTO causes alternate testing (at 5-second intervals) of both. Allowed values are AUTO A_CABLE B_CABLE

2.3.2 General Parameters

TESTS	Allowed values are 1 through 4
ERROR_LIMIT	Allowed values are 1 through 32767 or "-"
PASS_LIMIT	Allowed values are 1 through 32767 or "-"
REPORT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
INHIBIT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
SCALE	Allowed values are 0 through 999
MINUTES_LIMIT	Allowed values are 0 through 32767 or "-"
ABBREVIATION	Allowed values are 0 through 4

2.4 PRESET PARAMETER VALUES FOR AOMT

IOP_ADDRESS = 0

POINT_SLOT_MASK = 1-16,0

UNITS_OF_MEASURE = CURRENT

REF_100_PERCENT_VALUE = 20.0

REF_0_PERCENT_VALUE = 4.0

STARTING_VALUE = 4.0

MAXIMUM_VALUE = 20.0

MINIMUM_VALUE = 4.0

STEP_VALUE = 1.0

STEP_TIME = 1

MAX_REACHED_ACTION = REVERSE_STEP_SIGN

MIN_REACHED_ACTION = REVERSE_STEP_SIGN

START_REACHED_ACTION = NO_ACTION

ENABLE =

DISABLE = DEMAND

PATTERN = RANDOM

FTA_CABLE_SWITCH = AUTO

TESTS = 1-4

ERROR_LIMIT = 5

PASS_LIMIT = -

REPORT = LOG

INHIBIT = PASSNUMBER,TESTNUMBER,SUSPENDED

SCALE = 0

MINUTES_LIMIT = -

ABBREVIATION = 4

Down Stair Step:

STARTING_VALUE = same value as MINIMUM_VALUE

MAX_REACHED_ACTION = REVERSE_STEP_SIGN

MIN_REACHED_ACTION = GO_TO_MAX

START_REACHED_ACTION = GO_TO_MAX

Square Wave:

STARTING_VALUE = same value as MINIMUM_VALUE

MAX_REACHED_ACTION = GO_TO_MIN

MIN_REACHED_ACTION = GO_TO_MAX

START_REACHED_ACTION = GO_TO_MAX

Upper Square Wave:

STARTING_VALUE = value greater than MINIMUM_VALUE
and less than MAXIMUM_VALUE

MAX_REACHED_ACTION = GO_TO_START_VALUE

MIN_REACHED_ACTION = NO_ACTION

START_REACHED_ACTION = GO_TO_MAX

Lower Square Wave:

STARTING_VALUE = value greater than MINIMUM_VALUE
and less than MAXIMUM_VALUE

MAX_REACHED_ACTION = NO_ACTION

MIN_REACHED_ACTION = GO_TO_START_VALUE

START_REACHED_ACTION = GO_TO_MIN

Baseline with Pulse:

STARTING_VALUE = value greater than MINIMUM_VALUE
and less than MAXIMUM_VALUE

MAX_REACHED_ACTION = GO_TO_START_VALUE

MIN_REACHED_ACTION = GO_TO_START_VALUE

START_REACHED_ACTION = GO_TO_OTHER_MAX_OR_MIN

Sine Wave:

STARTING_VALUE = calculated by the program (halfway between
max and min values)

MAX_REACHED_ACTION = SINE_WAVE

MIN_REACHED_ACTION = SINE_WAVE

START_REACHED_ACTION = SINE_WAVE

STEP_TIME = specified value is used for sine-wave base cycle time

Test 02 Random Output SUB, MOD, EXER

Outputs a random or wig-wag (555/AAA) pattern as specified by the parameter PATTERN to all slots specified (by POINT_SLOT_MASK) of an AO IOP randomly selected from those specified by IOP_ADDRESS. It pauses for STEP_TIME and then resumes with a new random pattern or the complementing wig-wag pattern. Self-test failures are checked for after each output.

Test 03 Accuracy Check SUB, MOD

Outputs and holds a current to operator-selected slots until a new value is selected by the operator. When demand is enabled (ENABLE = DEMAND), this test outputs STARTING_VALUE to the AO IOP(s) of slots as specified by IOP_ADDRESS and POINT_SLOT_MASK and then makes DISABLE = DEMAND. This output value is held until demand is again enabled. Self-test failures are checked for every one second.

Test 04 Redundancy Test SUB, MOD

Outputs to the FTA are 100% (20 Ma) on Cable A and 0% (4 Ma) on Cable B. The test switches from IOP to IOP every 5 seconds if FTA_CABLE_SWITCH is AUTO. Otherwise, only output from the selected IOP (A_CABLE or B_CABLE) is evident.

Section 3 - DIGITAL INPUT MODULE TESTS (DIMT)

3.1 DIMT TEST ELEMENTS

This program tests Digital Input Modules and their associated FTAs that are connected to a selected PMM, APMM, HPMM.

3.2 USE INFORMATION

The tests in this program run on a single digital input IOP, if a value for IOP_ADDRESS is supplied, or on all configured DI IOPs, if that parameter value equals zero.

A pass for either of these tests consists of as many scans of all cards under test that can be accomplished in approximately one minute; therefore, the typical time required for a pass, even on a system of maximum configuration, remains about one minute.

NOTE

Because of memory constraints, DIMT and PIMT cannot run simultaneously.

3.3 SETUP PARAMETERS

3.3.1 DIMT-Specific Parameters

The following operator-selected values are used by one or more of the tests in this program.

IOP_ADDRESS File and Slot address of a digital input IOP to be tested (the IOP must appear in the configuration table). See headings 1.1.3 through 1.1.5 for information on IOP selection. A value of 000 for this parameter calls for test of all digital input IOPs contained in the configuration table.

TIME_BETWEEN_REPORTS Allowed values are 0 through 32767 (seconds). A value of 0 seconds causes reports to be produced each time a slot (point) value changes.

ENABLE or DISABLE Allows the operator to enable or disable specific functions associated with the execution of tests. The only allowed entry value is

DEMAND_LOG—when demand is enabled a report is produced; the program then disables demand.

3.3.2 General Parameters

TESTS	Allowed values are 1 through 2
ERROR_LIMIT	Allowed values are 1 through 32767 or "-"
PASS_LIMIT	Allowed values are 1 through 32767 or "-"
REPORT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
INHIBIT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
SCALE	Allowed values are 0 through 999
MINUTES_LIMIT	Allowed values are 0 through 32767 or "-"
ABBREVIATION	Allowed values are 0 through 4

3.4 PRESET PARAMETER VALUES FOR DIMIT

IOP_ADDRESS = 0

TIME_BETWEEN_REPORTS = 30

ENABLE =

DISABLE = DEMAND_LOG

TESTS = 1-2

ERROR_LIMIT = 5

PASS_LIMIT = -

REPORT = LOG

INHIBIT = PASSNUMBER,TESTNUMBER,SUSPENDED

SCALE = 0

MINUTES_LIMIT = -

ABBREVIATION = 4

3.5 MODE CHARACTERISTICS FOR DIMIT

<u>System Mode</u>	<u>Programs Running</u>	<u>Tests Running</u>	<u>Delay Between Tests</u>
Subsystem	Only one	Only one	None
Module	One or more	Only one	Random
Exerciser	One or more	Only one	Random

3.6 INDIVIDUAL DIMIT TESTS

(System Modes)

Test 01 Change Count

SUB, MOD, EXER

Scans inputs of the digital IOPs under test (see the parameter IOP_ADDRESS), counts the changes, and produces a "Count Report" periodically or on demand (see the parameters TIME_BETWEEN_REPORTS and ENABLE or DISABLE). The report is output automatically if the count is about to overflow. The Count Report consists of a display of the number of changes since the last periodic report for each point. After the report is displayed (except for a demand display) the counters are zeroed. Failures such as "No Response" are alarmed as they occur.

Test 02 Change Log

SUB,MOD

Scans inputs of the digital IOPs under test (see the parameter IOP_ADDRESS) and produces a "Point Value Report" periodically or on demand (see the parameters TIME_BETWEEN_REPORTS and ENABLE or DISABLE). The point value report consists of a display of the actual values (0,1) for each point.

Section 4 - DIGITAL OUTPUT MODULE TESTS (DOMT)

4.1 DOMT TEST ELEMENTS

This program tests Digital Output Modules and their associated FTAs that are connected to a selected PMM, APMM, HPMM.

4.2 USE INFORMATION

The tests in this program run on a single digital output IOP, if a value for IOP_ADDRESS is supplied, or on all configured DO IOPs, if that parameter value equals zero.

A pass for any of these tests consists of writing values to all specified points (card slots) on all specified available DO cards. The values are random or operator-specified.

The testing of redundant IOPs is controlled by the parameter FTA_CABLE_SWITCH. When this parameter is set to AUTO, switching is automatic. When it is set to A_CABLE or B_CABLE, only the IOP associated with the particular cable is tested.

4.3 SETUP PARAMETERS

4.3.1 DOMT-Specific Parameters

The following operator-selected values are used by one or more of the tests in this program.

IOP_ADDRESS	File and Slot address of a digital output IOP to be tested (the IOP must appear in the configuration table). See headings 1.1.3 through 1.1.5 for information on IOP addressing and configuration. A value of 000 for this parameter calls for the testing of all DO IOPs contained in the configuration table.
POINT_MASK	Points on the DO IOP that are to be tested by Test 3. There are 16 points on an DO IOP and each point to be tested must be specified. For example, the entry 1, 3, 5-12, 14, 0 indicates that point 1, point 3, points 5 through 12 and point 14 are to be tested. The terminal 0 indicates the end of test point input and must be included.
TIME	The pause time (in units of 100 milliseconds) between patterns if PATTERN is set to USER_DEFINED or the time between points if PATTERN is set to any other value. Allowed values are 0 to 200 (e.g. 10 = 1 second).
PULSE_TIME	The pulse time (in units of one second) when MODE value is MOMENTARY. Allowed values are 0.00 to 60.00 in 0.01 increments. (e.g. 0.04 = 40 milliseconds, 1.00 = 1 second). PULSE_TIME is used by Test 3 only.

PULSE_PERIOD	The pulse period (in units of one second) when MODE value is PULSE. Allowed values are 0.000 to 120.000. (e.g., 0.001 = 1 millisecond, 1.02 = 1.02 seconds). PULSE_PERIOD is used by Test 3 only.
PERCENT_PULSE	Percentage of PULSE_PERIOD that the pulse is on (when MODE value is PULSE). Allowed values are 0.0 to 100.0. PERCENT_PULSE is used by Test 3 only.
PATTERN	Selection of output pattern. Allowed values are - WALK0 (walking zeros) - WALK1 (walking ones) - MARCH0 (marching zeros) - MARCH1 (marching ones) - USER_DEFINED - WIG_WAG (\$AAAA/\$5555) - PING_PONG (marching ones, points 1-16, then 16-1)
DO_MODE	Mode of operation for the DOM IOP, establishing how it drives/stimulates its slots. Allowed values are - LATCHED - PULSED - MOMENTARY PULSED and MOMENTARY modes are effective for Test 3 only.
FTA_CABLE_SWITCH	Applies to IOP pairs only. Allows the test operator to select which IOP is to be tested by specifying the FTA cable associated with the IOP under test. Selection of AUTO causes alternate testing of both. Allowed values are - AUTO - A_CABLE - B_CABLE
USER_PATTERN	The output pattern to be used when PATTERN value is USER_DEFINED. Allowed values are \$0000 to \$FFFF.

4.3.2 General Parameters

TESTS	Allowed values are 1 through 3
ERROR_LIMIT	Allowed values are 1 through 32767 or "-"
PASS_LIMIT	Allowed values are 1 through 32767 or "-"
REPORT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
INHIBIT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
SCALE	Allowed values are 0 through 999
MINUTES_LIMIT	Allowed values are 0 through 32767 or "-"
ABBREVIATION	Allowed values are 0 through 4

4.4 PRESET PARAMETER VALUES FOR DOMT

IOP_ADDRESS = 0

POINT_MASK = 1-32,0

TIME = 1

PULSE_TIME = 1.0

PULSE_PERIOD = 10.0

PERCENT_PULSE = 50.0

PATTERN = WIG_WAG

DO_MODE = LATCHED

USER_PATTERN = \$0

TESTS = 1-3

ERROR_LIMIT = 5

PASS_LIMIT = -

REPORT = LOG

INHIBIT = PASSNUMBER,TESTNUMBER,SUSPENDED

SCALE = 0

MINUTES LIMIT = -

ABBREVIATION = 4

4.5 MODE CHARACTERISTICS FOR DOMT

<u>System Mode</u>	<u>Programs Running</u>	<u>Tests Running</u>	<u>Delay Between Tests</u>
Subsystem	Only one	Only one	None
Module	One or more	Only one	Random
Exerciser	One or more	Only one	Random

4.6 INDIVIDUAL DOMT TESTS

(System Modes)

Test 01 Fixed Patterns SUB, MOD

Outputs in LATCHED MODE patterns established by the choice of value for PATTERN to all DO IOPs specified by IOP_ADDRESS. The test pauses by the TIME value between patterns if PATTERN = WIG_WAG or USER_DEFINED or between each slot/point if another value is used for PATTERN.

Test 02 Random Patterns SUB, MOD, EXER

Outputs in LATCHED MODE patterns established by the choice of value for PATTERN to randomly selected DO IOPs from among those specified by IOP_ADDRESS. The test pauses by the TIME value between patterns if PATTERN = WIG_WAG or USER_DEFINED or between each slot/point if another value is used for PATTERN.

Test 03 Output Patterns SUB, MOD

Outputs in LATCHED, MOMENTARY, or PULSED MODE patterns established by the choice of value for PATTERN to all DO IOPs specified by the parameter IOP_ADDRESS. The test pauses by the TIME value between patterns if PATTERN = WIG_WAG or USER_DEFINED or between each slot/point if another value is used for PATTERN. Only slots/points specified by POINT_MASK are affected.

Test 3 use of pulsed mode causes some patterns that are difficult to recognize unless the pulse-period is less than pulse-time.

Section 5 - HIGH LEVEL ANALOG INPUT MODULE TESTS (HLAI)

5.1 HLAI TEST ELEMENTS

This program tests High Level Analog Input Modules and their associated FTAs that are connected to a selected PMM, APMM, HPMM.

5.2 USE INFORMATION

The tests in this program run on a single high-level analog input IOP if a value for IOP_ADDRESS is supplied, or on all configured HLAI IOPs if that parameter value equals zero. Only one test can run at a time and the tests do not sequence from one to the other.

A pass for either of these tests consists of scanning the event list and reporting any errors of all cards under test. In Test 1 there is a demand log option that prints the value of all points of all cards under test.

The testing of redundant IOPs is controlled by the parameter FTA_CABLE_SWITCH. When this parameter is set to AUTO, switching is automatic. When it is set to A_CABLE or B_CABLE, only that IOP is tested.

5.3 SETUP PARAMETERS

5.3.1 HLAI-Specific Parameters

The following operator-selected values are used by one or more of the tests in this program.

IOP_ADDRESS	File and Slot address of a high level analog input IOP to be tested (the IOP must appear in the configuration table). See headings 1.1.3 through 1.1.5 for information on IOP selection. A value of 000 for this parameter calls for test of all high-level analog IOPs contained in the configuration table.
POINT_MASK	Specifies which of the 16 Points/Slots are to be tested on the selected HLAI IOP(s). For example, the entry of 1, 3-5, 10-13, 0 selects points/slots 1, 3, 4, 5, 10, 11, 12 and 13 to be tested. The terminal 0 indicates end of the list and must be included.

UNITS_OF_MEASURE Establishes the engineering units of the process values displayed by test 1 and the units of the test input values entered for test 2 (parameters IN_1 through IN_6). Allowed values are

PERCENT

VOLTS

CURRENT

INPUT_RANGE Establishes the full scale range in engineering units for process values. Allowed values are

V0_5 (0 - 5 volt range)

MA4_20 (4 - 20 milliamp range)

MA10_50 (10 - 50 milliamp range)

IMPORTANT: The following table shows all valid combinations of INPUT_RANGE and UNITS_OF_MEASURE accepted by Test 1.

		UNITS OF MEASURE			
		Percent	Volts	Current	
I	R				
N	A	V0_5	OK	OK	N/A
P	N	MA4_20	OK	OK	OK
U	G	MA10_50	OK	OK	OK
T	E				

ACCURACY Establishes the tolerance (as a percentage of full scale) for the hi/lo alarm trip points used in test 2. Allowed values are

C25DEG ($\pm 0.075\%$ of full scale for 25°C)

C0_TO_50DEG ($\pm 0.3\%$ of full scale for 0 to 50°C)

C10_TO_60DEG ($\pm 0.4\%$ of full scale for -10 to 60°C)

ONE_PERCENT ($\pm 1.0\%$ of full scale)

ENABLE or DISABLE Allows the operator to enable or disable specific functions associated with the execution of tests. The only allowed entry value is

DEMAND_LOG—when demand log is enabled a report is produced by test 1; the program then disables demand.

IN_1 . . IN_6

Floating point values in volts, milliamps or percent in the form x.xxxxxx. These values must be consistent with the parameters INPUT_RANGE and UNITS_OF_MEASURE. Test 2 attempts to match these values with the actual inputs. To disable one of these values, enter a question mark ("?").

IMPORTANT: The following table shows the permitted input value ranges for all valid combinations of INPUT_RANGE and UNITS_OF_MEASURE.

		UNITS OF MEASURE			
		Percent	Volts	Current	
I	R				
N	A	V0_5	0.0—100.0	0.0—5.0	N/A
P	N	MA4_20	-25.0—100.0	0.0—5.0	0.0—20.0
U	G	MA10_50	-25.0—100.0	0.0—5.0	0.0—50.0
T	E				

FTA_CABLE_SWITCH

Applies to redundant IOP pairs only. Allows the test operator—by specifying the FTA cable position—to select which IOP of a redundant pair will be tested. Selection of AUTO causes alternate testing (at 5-second intervals) of both. Allowed values are: AUTO, A_CABLE, B_CABLE.

5.3.2 General Parameters

TESTS	Allowed values are 1 through 2
ERROR_LIMIT	Allowed values are 1 through 32767 or "-"
PASS_LIMIT	Allowed values are 1 through 32767 or "-"
REPORT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
INHIBIT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
SCALE	Allowed values are 0 through 999
MINUTES_LIMIT	Allowed values are 0 through 32767 or "-"
ABBREVIATION	Allowed values are 0 through 4

5.4 PRESET PARAMETER VALUES FOR HLA

IOP_ADDRESS = 0

POINT_MASK = 1-16,0

UNITS_OF MEASURE = VOLTS

INPUT_RANGE = V0_5

ACCURACY = C25DEG

ENABLE =

DISABLE = DEMAND_LOG

IN_1 = ?

IN_2 = ?

IN_3 = ?

IN_4 = ?

IN_5 = ?

IN_6 = ?

FTA_CABLE_SWITCH = AUTO

TESTS = 1,2

ERROR_LIMIT = 5

PASS_LIMIT = -

REPORT = LOG

INHIBIT = PASSNUMBER,TESTNUMBER,SUSPENDED

SCALE = 0

MINUTES_LIMIT = -

ABBREVIATION = 4

5.6 INDIVIDUAL HLAI TESTS

Reads all input points of selected HLAI IOP(s) and attempts to match each value with one of the parameter values IN_1 through IN_6. When matches are made (input value equal to IN_n value plus or minus ACCURACY value), the point is configured to alarm on Hi/Lo or "Accuracy" trip points (any deviations outside IN_n plus or minus ACCURACY) and on HiHi/LoLo or "Range" trip points (any deviations beyond INPUT_RANGE plus or minus ACCURACY). When no value match is found for an input point, the point is alarmed and only its Range trip point is set. Points with other failures such as "No Response" also are alarmed.

Section 6 - LOW LEVEL ANALOG INPUT MODULE TESTS (LLAI)

6.1 LLAI TEST ELEMENTS

This program tests Low Level Analog Input Modules and their associated FTAs that are connected to a selected PMM, APM, HPMM.

6.2 USE INFORMATION

The tests in this program run on a single low-level analog input IOP if a value for IOP_ADDRESS is supplied, or on all configured LLAI IOPs if that parameter value equals zero. Only one test can run at a time and the tests do not sequence from one to the other.

A pass for either of these tests consists of scanning the event list and reporting any errors of all cards under test. In Test 1 there is a demand log option that prints the value of all points of all cards under test.

6.3 SETUP PARAMETERS

6.3.1 LLAI-Specific Parameters

The following operator-selected values are used by one or more of the tests in this program.

IOP_ADDRESS	File and Slot address of a low-level analog input IOP to be tested (the IOP must appear in the configuration table). See headings 1.1.3 through 1.1.5 for information on IOP selection. A value of 000 for this parameter calls for test of all low-level analog IOPs contained in the configuration table.
POINT_MASK	Specifies which of the eight Points/Slots are to be tested on the selected LLAI IOP(s). For example, the entry of 1, 3-6, 0 selects points/slots 1, 3, 4, 5 and 6 to be tested. The terminal 0 indicates end of the list and must be included.
UNITS_OF MEASURE	Establishes the engineering units of the process values displayed by test 1 and the units of the test input values entered for test 2 (parameters IN_1 through IN_6). Allowed values are PERCENT VOLTS CURRENT TEMP (in degrees Celsius)

INPUT_RANGE

Establishes the full scale range in engineering units for process values. Allowed values are

MV0_100 (0 -100 millivolt range)

V0_5 (0 - 5 volt range)

MA4_20 (4 - 20 milliamp range)

MA10_50 (10 - 50 milliamp range)

AUTO (0 - 100 millivolt or 0 - 5 volt)

TC (thermocouple input)

IMPORTANT: The following table shows all valid combinations of INPUT_RANGE and UNITS_OF_MEASURE accepted by Test 1.

		UNITS OF MEASURE			
		Percent	Volts	Current	Temp
I	R	MV0_100	OK	OK	N/A
N	A	V0_5	OK	OK	N/A
P	N	MA4_20	OK	OK	OK
U	G	MA10_50	OK	OK	OK
T	E	AUTO	N/A	OK	N/A
		TC	N/A	N/A	OK

FILTER_DELAY

A floating point number in the range 0.0 to 60.0 that defines the PV Filter Lag Time in minutes (database parameter TF). A value of 0.0 defines no filtering and the program uses PVCALC as the process value. A value other than 0.0 enables filtering and PV is used as the process value. The program honors a filter delay other than 0.0 only if the UNITS_OF_MEASURE is TEMP and the INPUT_RANGE is TC.

ACCURACY

Establishes the tolerance (as a percentage of full scale) for the hi/lo alarm trip points used in test 2. Allowed values are

C25DEG ($\pm 0.075\%$ of full scale for 25°C)

C0_TO_50DEG ($\pm 0.25\%$ of full scale for 0 to 50°C)

C10_TO_60DEG ($\pm 0.32\%$ of full scale for -10 to 60°C)

ONE_PERCENT ($\pm 1.0\%$ of full scale)

ENABLE or DISABLE Allows the operator to enable or disable specific functions associated with the execution of tests. The only allowed entry values are

DEMAND_LOG—when demand log is enabled a report is produced by test 1; the program then disables demand.

OTD—Open Thermocouple Detection enable or disable.

THERMOCOUPLE Defines thermocouple type for the TC points/slots when UNITS_OF_MEASURE is TEMP. Allowed values are

JTHERM (-200.0 to 1200.0)

KTHERM (-200.0 to 1370.0)

ETHERM (-200.0 to 1000.0)

TTHERM (-230.0 to 400.0)

BTHERM (100.0 to 1820.0)

STHERM (0.0 to 1700.0)

RTHERM (0.0 to 1700.0)

RPTHERM (0.0 to 1700.0)

IN_1 . . IN_6 Floating point values in volts, milliamps, percent, or temp in the form x.xxxxxx. These values must be consistent with the parameters INPUT_RANGE and UNITS_OF_MEASURE. Test 2 attempts to match these values with the actual inputs. To disable one of these values, enter a question mark ("?").

IMPORTANT: The following table shows the input value ranges permitted by Test 2 for all valid combinations of INPUT_RANGE and UNITS_OF_MEASURE.

			UNITS OF MEASURE			
			Percent	Volts	Current	Temp
I R	MV0_100		0.0—100.0	0.0—0.1	N/A	N/A
N A	V0_5		0.0—100.0	0.0—5.0	N/A	N/A
P N	MA4_20		-25.0—100.0	0.0—5.0	0.0—20.0	N/A
U G	MA10_50		-25.0—100.0	0.0—5.0	0.0—50.0	N/A
T E	AUTO		N/A	0.0—5.0	N/A	N/A
	TC		N/A	N/A	N/A	*

*TC ranges vary by type; see THERMOCOUPLE parameter.

FREQ50_60 Identifies the input ac line frequency, FIFTY or SIXTY Hz.

6.3.2 General Parameters

TESTS	Allowed values are 1 through 2
ERROR_LIMIT	Allowed values are 1 through 32767 or "-"
PASS_LIMIT	Allowed values are 1 through 32767 or "-"
REPORT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
INHIBIT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
SCALE	Allowed values are 0 through 999
MINUTES	Allowed values are 0 through 32767 or "-"
ABBREVIATION	Allowed values are 0 through 4

6.4 PRESET PARAMETER VALUES FOR LLA

IOP_ADDRESS = 0

POINT_MASK = 1-8,0

UNITS_OF MEASURE = VOLTS

INPUT_RANGE = MV0_100

FILTER_DELAY = ?

ACCURACY = C25DEG

ENABLE =

DISABLE = DEMAND_LOG,OTD

THERMOCOUPLE = JTHERM

IN_1 = ?

IN_2 = ?

IN_3 = ?

IN_4 = ?

IN_5 = ?

IN_6 = ?

FREQ50_60 = SIXTY

TESTS = 1-2

ERROR_LIMIT = 5

PASS_LIMIT = -

REPORT = LOG

INHIBIT = PASSNUMBER,TESTNUMBER,SUSPENDED

SCALE = 0

MINUTES = -

ABBREVIATION = 4

6.5 MODE CHARACTERISTICS FOR LLAI

<u>System Mode</u>	<u>Programs Running</u>	<u>Tests Running</u>	<u>Delay Between Tests</u>
Subsystem	Only one	Only one	None
Module	One or more	Only one	Random
Exerciser	One or more	Only one	Random

6.6 INDIVIDUAL LLA I TESTS

(System Modes)

Test 01 Range Check

SUB, MOD

Reads all input points of selected LLA I IOP(s) and alarms any point values that fall outside the INPUT_RANGE plus or minus ACCURACY. Points with other failures such as "No Response" also are alarmed. When DEMAND_LOG is enabled, all point values of selected LLA I IOPs are logged in terms of the assigned UNIT_OF_MEASURE.

Exception 1: When INPUT_RANGE = AUTO, individual points are configured first to the MV0-100 range; then, if the value read is not within that range plus or minus the value for ACCURACY, the value is set to the V0_5 range. Points with a BadPv are configured as V0_5.

Exception 2: When INPUT_RANGE = TC, the points have NO tolerance and the range is set to the extended range of the thermocouple type specified in the THERMOCOUPLE parameter.

Test 02 Accuracy Check

SUB, MOD, EXER

Reads all input points of selected LLA I IOP(s) and attempts to match each value with one of the parameter values IN_1 through IN_6. When matches are made (input value equal to IN_n value plus or minus ACCURACY value) the point is then configured to alarm on Hi/Lo or "Accuracy" trip points (any deviations outside IN_n plus or minus ACCURACY) and on HiHi/LoLo or "Range" trip points (any deviations beyond INPUT_RANGE plus or minus ACCURACY). When no value match is found for an input point, the point is alarmed and only its Range trip point is set. Points with other failures such as "No Response" also are alarmed.

Exception 1: When INPUT_RANGE = AUTO, individual points are configured first to the MV0-100 range; then, if the value read is not within that range plus or minus the value for ACCURACY, the value is set to the V0_5 range. Points with a BadPv are configured as V0_5. After determining the range, the test tries to match each point against the IN_n parameter values and set accuracy limits as above. The disadvantage of selecting AUTO as INPUT_RANGE is the configuration overhead and the possibility of generating some "soft fail" recovery events that can show up later in the test.

Exception 2: When INPUT_RANGE = TC, the range is set to the extended range limits of the thermocouple type specified in the THERMOCOUPLE parameter with no tolerance. Accuracy limits (IN_n plus or minus ACCURACY) are held within the extended range limits of the specified thermocouple type.

Section 7 - SMART TRANSMITTER INTERFACE MODULE TESTS (STIM)

7.1 STIM TEST ELEMENTS

This program is used to verify or monitor the operation of Smart Transmitter Interface Modules (STI) and their associated FTAs that are connected to a selected PMM, APMM, HPMM.

7.2 USE INFORMATION

This program requires that each STI FTA point to be tested must be connected to a Smart Transmitter (similar to the Model 600 "0-400 Inches of Water" Smart Pressure Transmitter) that is capable of producing the required constant pressure indication of 100 inches of water. Tests 1 and 2 require the same model transmitter to be used throughout; however, not all points on all STI IOPs need be connected to transmitters (see the parameters PT_CONFIG and POINT_MASK).

Test Initialization occurs each time the RUN command is issued and consists of

1. Building an internal list of the STI IOPs and the implemented points on each IOP to be tested, based on the parameters IOP_ADDRESS and PT_CONFIG. Any failed STI IOPs are dropped from this list as they fail. If one of a redundant pair of IOPs fails, both are dropped from the list.
2. Checking value parameters for reasonableness before their use.

Not all points on the STI IOPs need to be connected to transmitters, but those that are not need to be skipped over through your selection of the PT_CONFIG and POINT_MASK parameter values.

Only one STIM test can run at a time.

Tests 1, 2, and 3 run on the IOP specified by IOP_ADDRESS or on all configured STI IOPs when the IOP_ADDRESS parameter value equals zero. Test 3 is further restricted to run only on a single point as specified by the parameter POINT_MASK.

Tests 1 and 3 require operator intervention.

Note that cable switching, whether automatic or demanded, takes one minute each time. If automatic switching is specified, it adds one minute per pass.

7.3 SETUP PARAMETERS

7.3.1 STIM-Specific Parameters

The following operator-selected values are used by one or more of the tests in this program.

IOP_ADDRESS	File and Slot address of an STI IOP to be tested (the IOP must appear in the configuration table). See headings 1.1.3 through 1.1.5 for information on IOP selection. A value of 0 for this parameter calls for test of all STI IOPs contained in the configuration table. (For redundant STIMs, also see the parameter FTA_CABLE_SWITCH.)
UNIT_NAME	The name (1-20 characters) of the engineering units associated with PVCALC.
DISPLAY_TYPE	The allowed entry values are

VALUES—Specifies the display of

- IOP address
- point number
- point status (STATE)
- Upper and Lower Range limits (URL/LRL)
- Upper and Lower Range values (URV/LRV)
- input value as a percent of range (PVRAW)
- input value in engineering units (PVCALC)
- engineering units name as specified by UNIT_NAME

ID_PLUS_VALUES—Specifies the display of

- IOP address
- point number
- transmitter ID (from SENSORTYP)
- transmitter serial number (SERIALNO)
- STI tag name (SFITAG)
- point status (STATE)
- Upper and Lower Range limits (URL/LRL)
- Upper and Lower Range values (URV/LRV)
- input value as a percent of range (PVRW)
- input value in engineering units (PVCALC)
- engineering units name as specified by UNIT_NAME

DETAILED—Specifies display of the

ID_PLUS_VALUES log plus the following:

- PV characterization (PVCHAR)
- filter option (DAMPING)
- cold junction status (CJTACT)
- open thermocouple detection status (PIUOTDCF)
- process value status (PVSTS)
- secondary variable (SECVAR)
- digitally enhanced configuration mode (DECONF)

a 127-character ASCII string (S1 parameter) containing:

- 16 characters of transmitter scratch pad 1
- 16 characters of transmitter scratch pad 2
- four 16-character fields showing the four highest priority detailed transmitter status conditions
- 31 character positions reserved for future use

PT_CONFIG	The allowed entry values are USE_POINT_MASK—Used if all STI IOPs have the same points/slots connected to a smart transmitter. AUTO—Used when the program will determine which points/slots on each STI IOP are connected to a smart transmitter.
POINT_MASK	Defines the points on the STI IOP(s) to be tested. To test all 16 points, enter 1-16,0. To test a partial set such as point 2, points 5 through 7 and point 14, enter 2,5-7,14,0. Note that the terminating 0 must be included.
HIVALUE	A floating point number—used by Test 1 only—which defines the Upper Range Value (URV), or 100% value of the desired operating range. Note that all defined points on all defined STI IOPs will be affected.
LOVALUE	A floating point number—used by Test 1 only—that defines the Lower Range Value (LRV), or 0% value of the desired operating range. Note that all defined points on all defined STI IOPs will be affected.
ENTRY	The entry point for alphanumeric information entered interactively during Test 3.
ENABLE/DISABLE	Allows the operator to enable or disable specific functions associated with the execution of tests. The allowed entry values for STIM are DEMAND_LOG—When demand log is enabled, a report is produced by test 1 or test 2; the program then disables demand log. The log includes all the data specified by DISPLAY_TYPE for all points specified by IOP_ADDRESS and PT_CONFIG. LOAD_XMTR—When enabled, Test 1 sends the range values in HIVALUE and LOVALUE to all transmitters specified by IOP_ADDRESS and PT_CONFIG. This parameter is disabled after the range values are transmitted, so the operator has to re-enable LOAD_XMTR each time a new set of values is to be sent to the transmitters. Also, when you use FTA_CABLE_SWITCH to force a swap to a redundant partner, LOAD_XMTR must be enabled to update the partner's PVRAW and PVCALC values.
FREQ50_60	Identifies the frequency of the input AC line voltage; FIFTY or SIXTY Hz.

FTA_CABLE_SWITCH	Identifies which IOP of a redundant pair to test. The allowed values are A_CABLE, B_CABLE, and AUTO_ . See the individual test explanations at heading 7.6 for use restrictions.
STIX_INSTALLED	Allows the operator to use the factory-built simulator equipment to test with a significant performance improvement. The allowed values are NO and YES.

7.3.2 General Parameters

TESTS	Allowed values are 1 through 3
ERROR_LIMIT	Allowed values are 1 through 32767 or "-"
PASS_LIMIT	Allowed values are 1 through 32767 or "-"
REPORT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
INHIBIT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
SCALE	Allowed values are 0 through 999
MINUTES	Allowed values are 0 through 32767 or "-"
ABBREVIATION	Allowed values are 0 through 4

7.4 PRESET PARAMETER VALUES FOR STIM

IOP_ADDRESS = 0

UNIT_NAME = INCHES OF WATER

DISPLAY TYPE = VALUES

PT_CONFIG = USE_POINT_MASK

POINT_MASK = 1-16,0

HIVALUE = 0

LOVALUE = 0

ENTRY =

ENABLE =

DISABLE = DEMAND_LOG,LOAD_XMTR

FREQ50_60 = SIXTY

FTA_CABLE_SWITCH = A_CABLE

STIX_INSTALLED = NO

TESTS = 1-3

ERROR_LIMIT = 5

PASS_LIMIT = -

REPORT = LOG

INHIBIT = PASSNUMBER,TESTNUMBER,SUSPENDED

SCALE = 0

MINUTES = -

ABBREVIATION = 4

7.5 MODE CHARACTERISTICS FOR STIM

<u>System Mode</u>	<u>Programs Running</u>	<u>Tests Running</u>	<u>Delay Between Tests</u>
Subsystem	Only one	Only one	None
Module	One or more	Only one	Random
Exerciser	One or more	Only one	Random

Test 01	Manual Range and Value Check	SUB, MOD
---------	------------------------------	----------

The operator changes the operating range (span) of the transmitter(s)—thereby changing the "percent of range" value (PVRW) returned to the STI point—by changing the value of parameters HIVALUE and LOVALUE and then enabling LOAD_XMTR. Following command execution, the STI database for each tested point is displayed as specified by the parameter DISPLAY_TYPE. The demand log can be enabled at any time in this test.

Value of the parameter FTA_CABLE_SWITCH cannot be AUTO_ for this test. It can be changed between A_CABLE and B_CABLE by the operator at any time; the switch is made at beginning of the next pass. When switching cables, you also must enable LOAD_XMTR in order to update the new IOP database, with current PVRAW and PVCALC values.

Test 02 Auto Range and Value Test SUB, MOD, EXER

For each pass, the program selects one STI IOP at random to test. All specified points on this IOP are affected (see the parameter PT_CONFIG). The operating range (span) is selected at random, the point database is updated and then downloaded to the transmitter(s). The sensor reading for each point is then read from the point database in percent of range (PVRAW) and verified.

When FTA_CABLE_SWITCH is set to A_CABLE, or to B_CABLE, then only IOPs on the selected cable are tested. When AUTO_ is specified, the test switches between cables A and B at the end of each pass. The value of FTA_CABLE_SWITCH can be changed at any time, but the effect is not seen until the next pass.

Test 03 Utility Test

SUB, MOD

Allows the operator to display and/or redefine a transmitter's database through an interactive display. Content of the display is dependent on the parameter DISPLAY_TYPE. The operator makes selections in response to a menu of requests showing the following parameters:

URV—upper range value

LRV—lower range value

DAMPING—first order filtering option in the transmitter

UPLOAD—upload the data base from the transmitter

DOWNLOAD—download the database to the transmitter

SFITAG—STI tag name

DISPLAY THE LOG—forces a display of the log specified by the setup parameter DISPLAY_TYPE

ACTIVATE—set point execution state to active (must be the final command entered in a sequence of commands)

After either URV or LRV value entry, you are prompted to also enter a value for TOLERANCE (used to set the high or low trip point).

Example sequence showing how to set URV:

<u>Command Choice</u>	<u>Prompt</u>	<u>Your response</u>
URV	UR Value	Real number input
	Tolerance	Real number input
DOWNLOAD	none	none
ACTIVATE		

For this test to run, PT_CONFIG must be set to USE_POINT_MASK and POINT_MASK must specify only one point.

Value of the parameter FTA_CABLE_SWITCH cannot be AUTO_ for this test. A change of this parameter value (between A_CABLE and B_CABLE) while the test is running does not take effect until the next RUN command.

Section 8 - PULSE INPUT MODULE TESTS (PIMT)

8.1 PIMT TEST ELEMENTS

This program tests Pulse Input Modules (PIMs) and their associated FTAs that are connected to a selected PMM, APMM, HPMM. An external signal source is required to supply test input.

8.2 USE INFORMATION

The tests in this program run on a single Pulse Input IOP module if a value for IOP_ADDRESS is supplied, or on all configured pulse input IOPs if that parameter value equals zero. Only one test can run at a time and the tests do not sequence from one to the other.

A pass for either of these tests consists of one collection for each point in the point mask, from each PIM under test.

NOTE

Because of memory constraints, DIMIT and PIMT tests cannot run simultaneously.
--

8.3 SETUP PARAMETERS

8.3.1 PIMT-Specific Parameters

The following operator-selected values are used by one or more of the tests in this program.

IOP_ADDRESS	File and Slot address of a pulse input IOP module to be tested (the IOP must appear in the configuration table). See headings 1.1.3 through 1.1.5 for information on IOP selection. A value of zero (0) for this parameter calls for test of all pulse input IOPs contained in the configuration table.
TIME_BETWEEN_REPORTS	Interval between reports in seconds. An entry of zero (0) means no reports are to be issued.
REPORT_TYPE	Type of report to be produced by Test 1—TOTAL_COUNT or PULSE_RATE. See headings 8.7.1 and 8.7.2.
ENABLE/DISABLE	Allows the operator to enable or disable specific functions associated with the execution of tests. The only allowed entry value is DEMAND_LOG—when demand log is enabled, Test 1 produces the report specified by the REPORT_TYPE parameter, or Test 2 produces an "Out of Tolerance" report. The program then disables demand.
POINT_MASK	Specifies which of the eight points/slots are to be tested on the selected PIMT IOP(s). For example, the entry of 1, 3-6, 0 selects points/slots 1, 3, 4, 5 and 6 to be tested. The terminal 0 indicates end of the list and must be included.
RATE_n_EXPECTED	Specifies the test 2 Target and Tolerance values for each of up to eight possible pulse input rates. For example, if RATE_3_EXPECTED = 1000,2, the program accepts a rate of 1000 pps (plus or minus 2 pps) as one of the possible pulse input rates. The Target plus or minus the Tolerance must be greater than or equal to zero and less than or equal to 50,000.

8.3.2 General Parameters

TESTS	Allowed values are 1 through 2
ERROR_LIMIT	Allowed values are 1 through 32767 or "-"
PASS_LIMIT	Allowed values are 1 through 32767 or "-"
REPORT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
INHIBIT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
SCALE	Allowed values are 0 through 999
MINUTES	Allowed values are 0 through 32767 or "-"
ABBREVIATION	Allowed values are 0 through 4

8.4 PRESET PARAMETER VALUES FOR PIMT

IOP_ADDRESS = 0

TIME_BETWEEN_REPORTS = 10

REPORT_TYPE = PULSE_RATE

ENABLE = <blank>

DISABLE = DEMAND_LOG

POINT_MASK = 1-8,0

RATE_1_EXPECTED = 10000,50

RATE_2_EXPECTED = 20000,100

RATE_3_EXPECTED = 1000,5

RATE_4_EXPECTED = 0,0

RATE_5_EXPECTED = 0,0

RATE_6_EXPECTED = 0,0

RATE_7_EXPECTED = 0,0

RATE_8_EXPECTED = 0,0

TESTS = 1-2

ERROR_LIMIT = 5

PASS_LIMIT = -

REPORT = LOG

INHIBIT = PASSNUMBER,TESTNUMBER,SUSPENDED

SCALE = 0

MINUTES = -

ABBREVIATION = 4

8.5 MODE CHARACTERISTICS FOR PIMT

<u>System Mode</u>	<u>Programs Running</u>	<u>Tests Running</u>	<u>Delay Between Tests</u>
Subsystem	Only one	Only one	None
Module	One or more	Only one	Random
Exerciser	One or more	Only one	Random

8.6 INDIVIDUAL PIMT TESTS

(System Modes)

Test 01 Report

SUB, MOD

This test collects either the pulse rate or total count (see the parameter REPORT_TYPE) for each selected point (see the parameter POINT_MASK) from the PIMs under test (see the parameter IOP_ADDRESS) and reports the value to the operator. The value collected is reported periodically or on demand (see the parameters TIME_BETWEEN_REPORTS and ENABLE/DISABLE).

This test also will alarm the selected points with failures such as "No Response."

This is a Utility test and thus must be the only test running in the node.

Test 02 Accuracy

SUB, MOD, EXER

This test first attempts to match the pulse rate for each selected point (see the parameters IOP_ADDRESS and POINT_MASK) with the values entered in the RATE_n_EXPECTED parameters. Depending on the results of this comparison, one of the following actions is taken:

- When a pulse rate is within one of the eight rate-expected ranges (target value plus or minus its tolerance), high and low trip values are set for the point at the upper and lower limits of the expected range.
- When a pulse rate fails to fall within one of the expected ranges, the operator is notified that no match exists for that point and is provided the actual point pulse rate through the Point No Match report (see heading 8.7.3). The point's out of tolerance count value is set to 32767, trip values are not set, and the point is not alarmed during subsequent passes.

Subsequent passes of this test will monitor and report to the operator any Point Out of Tolerance Alarms (see heading 8.7.4) and Point Return to Within Tolerance Events (see heading 8.7.5).

When the ENABLE parameter is set to DEMAND_LOG, this test reports the total number of out of tolerance alarms collected for each point since the beginning of the test by issuing an Out of Tolerance Report (see heading 8.7.6). This same report is issued automatically whenever the count for one of the points reaches the overflow limit of 32767. (Whenever a point reaches this limit no further alarms are recorded for that point.)

This test also will alarm selected points with failures such as "No Response."

8.7 PIMT REPORTS AND ALARMS

Following are samples of the reports and alarms generated by Tests 1 and 2.

8.7.1 Total Count Report

A Total Count Report (Test 1) consists of a display of the count at the start of test, last count, and the net count (delta between last count and start of test count) for each selected channel for each PIM under test. When a channel is not selected, "N/A" is displayed instead of counts.

DD-MMM-YYYY HH:MM:SS NMA nn COMM nn - PULSE INPUT COUNT REPORT

PI IOP FSS

Pt	START	LAST	NET	Pt	START	LAST	NET
1	nnnnnnnnnn	nnnnnnnnnn	nnnnnnnnnn	2	N/A	N/A	N/A
3	nnnnnnnnnn	nnnnnnnnnn	nnnnnnnnnn	4	nnnnnnnnnn	nnnnnnnnnn	nnnnnnnnnn
5	N/A	N/A	N/A	6	N/A	N/A	N/A
7	nnnnnnnnnn	nnnnnnnnnn	nnnnnnnnnn	8	N/A	N/A	N/A

8.7.2 Pulse Rate Report

A Pulse Rate Report (Test 1) consists of a display of the pulse rate for each selected channel for each PIM under test. When a channel is not selected, "N/A" is displayed instead of a pulse rate.

DD-MMM-YYYY HH:MM:SS NMA nn COMM nn - PULSE INPUT RATE REPORT

VALUES IN PULSES PER SECOND

		POINT_MASK							
PI	IOP	1	2	3	4	5	6	7	8
-----+-----+-----+-----+-----+-----+-----+-----+-----									
Fss		nnnnn	N/A	nnnnn	nnnnn	N/A	N/A	nnnnn	N/A
Fss		nnnnn	N/A	nnnnn	nnnnn	N/A	N/A	nnnnn	N/A
*					*				*
*					*				*
*					*				*
Fss		nnnnn	N/A	nnnnn	nnnnn	N/A	N/A	nnnnn	N/A

8.7.3 Point No Match Report

A Point No Match Report (Test 2) consists of the identity of the point and its actual point pulse rate.

DD-MMM-YYYY HH:MM:SS NMA nn COMM nn

IOP Fss

Point n - Pulse rate doesn't match any Rate_n_Expected

Pulse Rate Read = nnnnn

8.7.4 Point Out of Tolerance Alarm

A Point out of Tolerance Alarm (Test 2) consists of the identity of the point, the event record contents, the alarm number, the "loop" and "pass" counts, and the alarm message.

```
DD-MMM-YYYY HH:MM:SS  NMA nn    COMM nn    PIMT n    Alarm nnn    IOP Fss
ENTRY IN THE EVENT LIST                                LOOP nnn    PASS nnnn
Event Record contents = $HH $HH $HH $HH $HH $HH $HH $HH $HH $HH $HH
Point n had an "Alarm" Event; PV is BELOW LOW LOW ALARM TRIP POINT
Current PV  nnnnn.nnnn          Crossed TP  nnnnn.nnnn
```

8.7.5 Point Return to Within Tolerance Report

A Point Return to Within Tolerance Report (Test 2) consists of the identity of the point, the event record contents, the alarm number, the "loop" and "pass" counts, and the "returned from alarm" message.

```
DD-MMM-YYYY HH:MM:SS  NMA nn    COMM nn    PIMT n    Alarm nnn    IOP Fss
ENTRY IN THE EVENT LIST                                LOOP nnn    PASS nnnn
Event Record contents = $HH $HH $HH $HH $HH $HH $HH $HH $HH $HH $HH
Point n RECOVERED FROM "Alarm"; PV is BELOW LOW LOW ALARM TRIP POINT
Current PV  nnnnn.nnnn          Crossed TP  nnnnn.nnnn
```

8.7.6 Out of Tolerance Report

The Out of Tolerance Report consists of the number of out of tolerance values read for each selected point from each PIM under test. If a point is not selected "N/A" is displayed in place of the number of out of tolerance values. Note that an out of tolerance value of 32767 can mean either that the point did not match an expected rate or that the point had a very abnormal tolerance error alarm (and return) rate.

DD-MMM-YYYY HH:MM:SS NMA nn COMM nn - OUT OF TOLERANCE REPORT

COUNT SINCE START OF TEST

	POINT_MASK							
PI IOP	1	2	3	4	5	6	7	8
FSS	nnnnn	N/A	nnnnn	nnnnn	N/A	N/A	nnnnn	N/A
FSS	nnnnn	N/A	nnnnn	nnnnn	N/A	N/A	nnnnn	N/A
*				*				*
*				*				*
*				*				*
FSS	nnnnn	N/A	nnnnn	nnnnn	N/A	N/A	nnnnn	N/A

Section 9 - LOW LEVEL MUX ANALOG INPUT MODULE TESTS (LLMX)

9.1 LLMX TEST ELEMENTS

This program tests Low Level Mux Analog Input Modules and their associated FTAs that are connected to a selected PMM, APMM, HPMM. External signal source(s) appropriate to the FTA type(s) are required to supply test input(s).

9.2 USE INFORMATION

The tests in this program run on a single low-level Mux analog input IOP if a value for IOP_ADDRESS is supplied, or on all configured LLMX IOPs if that parameter value equals zero. Only one test can run at a time and the tests do not sequence from one to the other.

The test program checks the type of each attached FTA, and in some instances will alarm mismatches between the FTA type you specify and the actual hardware it finds. When you select a specific IOP or IOPs to be tested and also select only one FTA type and the program finds a selected point to be of the other FTA type, it will alarm. If, however, you have used a zero IOP_ADDRESS entry and/or have an entry of BOTH in the parameter FTA_TYPES_TO_TEST there can be no mismatch because either FTA type can be expected at any IOP.

If COLD_RESTART is disabled, no cold restart will be allowed. The test will configure and activate the points more than once before alarming the events. In essence, any event or alarm caused by a cold restart will be discarded.

A pass for either of these tests consists of scanning the event list and reporting any errors of all cards under test. In Test 1 there is a demand log option that logs the value of all selected points of all cards under test.

9.3 SETUP PARAMETERS

9.3.1 LLMX-Specific Parameters

The following operator-selected values are used by one or more of the tests in this program.

IOP_ADDRESS	File and Slot address of a low-level Mux analog input IOP to be tested (the IOP must appear in the configuration table). See headings 1.1.3 through 1.1.5 for information on IOP selection. A value of zero (0) for this parameter calls for test of all low-level analog Mux IOPs contained in the configuration table.
POINT_MASK	Specifies which of the 32 points/slots on the selected LLMX IOP(s) are to be tested. For example, the entry of 1, 3-6, 0 selects points/slots 1, 3, 4, 5, and 6 to be tested. The terminal 0 indicates end of the list and must be included.
UNITS_OF_MEASURE	Establishes for the TC or 100 Mv FTA points/slots, the engineering units of the process values displayed by Test 1 and the units of the test input values entered for Test 2 in parameters IN_1 through IN_6. (RTD FTA point/slot values are expressed in degrees C.) Allowed values are MVDC (millivolts dc) TEMP (TC inputs expressed as temperatures in °C)
THERMOCOUPLE	Defines which thermocouple type the TC points/slots are to be configured as when UNITS_OF_MEASURE is TEMP. Allowed values are JTHERM KTHERM ETHERM TTHERM BTHERM STHERM RTHERM RPTHERM

ENABLE or DISABLE	Allows the operator to enable or disable specific functions associated with the execution of tests. The only allowed entry values are DEMAND_LOG—when demand log is enabled, a report is produced by test 1; the program then disables demand. OTD—Open Thermocouple Detection enable or disable. COLD_RESTART—Cold Restart enable or disable
FILTER_DELAY	A floating point number in the range 0.0 to 60.0, which defines the PV Filter Lag Time in minutes (database parameter TF). A value of 0.0 defines no filtering and the program uses PVCALC as the process value. A value other than 0.0 enables filtering and PV is used as the process value. The program honors a filter delay other than 0.0 only if the UNITS_OF_MEASURE is TEMP.
ACCURACY	Allowed values are ONE_PERCENT (+/- 1.0% full scale) C25DEG (+/- 0.075% full scale for 25 °C) C0_TO_50DEG (+/- 0.25% full scale for 0 to 50 °C) C10_TO_60DEG (+/- 0.32% full scale for minus 10 to plus 60 °C)
FTA_TYPES_TO_TEST	Allowed values are RTD TC_0_100MVDC BOTH

IN_1 .. IN_6

Expected values (in floating point) for inputs from 0-100 mVdc or TC type FTAs. Values are expressed in terms of millivolts when UNITS_OF_MEASURE is MVDC, and in degrees C when UNITS_OF_MEASURE is TEMP. Values must fall within the following ranges

- 20.0 to 100.0 (millivolts)
- 200.0 to 1200.0 (JTHERM)
- 200.0 to 1370.0 (KTHERM)
- 200.0 to 1000.0 (ETHERM)
- 230.0 to 400.0 (TTHERM)
- 100.0 to 1200.0 (BTHERM)
- 0.0 to 1700.0 (STHERM)
- 0.0 to 1700.0 (RTHERM)
- 0.0 to 1700.0 (RPTHERM)

Failure to make a value entry for at least one of these parameters when FTA_TYPES_TO_TEST = TC_0_100 mVdc or BOTH will cause Test 2 to issue an error message and stop.

RTD_1 .. RTD_6

Expected values (in floating point) for inputs from RTD type FTAs. Allowed values are

- 200 to 850 (degrees C)

The default value is "?".

Failure to make a value entry for at least one of these parameters when FTA_TYPES_TO_TEST = RTD or BOTH will cause Test 2 to issue an error message and stop.

FREQ50_60

Identifies the input ac line frequency. Allowed values are:

- SIXTY
- FIFTY

9.3.2 General Parameters

TESTS	Allowed values are 1 through 2
ERROR_LIMIT	Allowed values are 1 through 32767 or "-"
PASS_LIMIT	Allowed values are 1 through 32767 or "-"
REPORT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED,
INHIBIT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED
SCALE	Allowed values are 0 through 999
MINUTES	Allowed values are 0 through 32767 or "-"
ABBREVIATION	Allowed values are 0 through 4

9.4 PRESET PARAMETER VALUES FOR LLMX

IOP_ADDRESS = 0

POINT_MASK = 1-32, 0

UNITS_OF_MEASURE = MVDC

THERMOCOUPLE = JTHERM

ENABLE = COLD_RESTART

DISABLE = DEMAND_LOG, OTD, NOT_USED

FILTER_DELAY = ?

ACCURACY = C25DEG

FTA_TYPES_TO_TEST = BOTH

IN_1 = ?

IN_2 = ?

IN_3 = ?

IN_4 = ?

IN_5 = ?

IN_6 = ?

RTD_1 = ?

RTD_2 = ?

RTD_3 = ?

RTD_4 = ?

RTD_5 = ?

RTD_6 = ?

FREQ50_60 = SIXTY

TESTS = 1-2

ERROR_LIMIT = 5

PASS_LIMIT = -

REPORT = LOG

INHIBIT = PASSNUMBER,TESTNUMBER,SUSPENDED

SCALE = 100

MINUTES_LIMIT = -

ABBREVIATION = 4

9.5 MODE CHARACTERISTICS FOR LLMX

<u>System Mode</u>	<u>Programs Running</u>	<u>Tests Running</u>	<u>Delay Between Tests</u>
Subsystem	Only one	Only one	None
Module	One or more	Only one	Random
Exerciser	One or more	Only one	Random

9.6 INDIVIDUAL LLMX TESTS

(System Modes)

Test 01 Range Check SUB, MOD

Reads all input points of selected LLMX IOP(s) and alarms any selected point values that fall outside range trip points. Points with other failures such as "No Response" also are alarmed.

For 0-100 mVdc or TC type FTAs:

- If UNITS_OF_MEASURE is MVDC, the range is -20 to 100 MVDC +/- ACCURACY.
- If UNITS_OF_MEASURE is TEMP, the range is set to the extended temperature range for the thermocouple type specified in the THERMOCOUPLE parameter.

For RTD FTAs, the range is -180 to +800 degrees C +/- ACCURACY (100 ohm RTD is assumed) when the IOP has a firmware revision less than 4.0. If it is 4.0 or greater, the range is -200 to +850 degree C.

When DEMAND is enabled, Test 1 scans and logs all selected point values of the selected LLMX IOPs.

For 0-100 mVdc or TC type FTAs:

- If UNITS_OF_MEASURE is MVDC, the input values are read from PVCALC and are expressed in millivolts.
- If UNITS_OF_MEASURE is TEMP and FILTER_DELAY is 0.0, the input values are read from PVCALC and are expressed in degrees C.
- If UNITS_OF_MEASURE is TEMP and FILTER_DELAY is **not** 0.0, the input values are read from PV and are expressed in degrees C.

For RTD type inputs, the values are expressed in degrees C.

Those FTAs not selected in FTA_TYPES_TO_TEST do not show up in the log. For the selected FTAs, those points not selected in POINT_MASK are shown as N/S in the log.

Test 02 Accuracy Check SUB, MOD, EXER

Accuracy check verifies that each selected IOP point matches one of the specified values (IN_n or RTD_n) within a tolerance range determined by ACCURACY. When a match is made, the point is configured to alarm on any deviations outside that range, and on deviations beyond range trip points for the FTA type and UNITS_OF_MEASURE selected (see Test 1). Log or alarm values are expressed in millivolts or degrees C for 0-100 mVdc/TC type FTAs, depending on UNITS_OF_MEASURE, and in DEG C for RTD type FTAs. When no value match is found, that point is alarmed and only its Range Trippoint is set. Points with other failures such as "No Response" also are alarmed.

The resistance values listed in the table that follows produce the indicated temperature values for RTD points.

Ohms	• C
50.06	-125
100.00	0
200.21	267
300.00	558
374.91	798
379.97	815

Section 10 - INPUT/OUTPUT PROCESSOR MONITOR (IOPM)

10.1 IOPM TEST ELEMENTS

This program monitors all selected IOPs in the Idle State for Events and their associated FTAs that are connected to a selected PMM, APMM, HPMM.

10.2 USE INFORMATION

The test in this program runs on all selected IOPs in the System List.

A pass for this test consists of one scan of the event list for all cards under test.

10.3 SETUP PARAMETERS

10.3.1 IOPM-Specific Parameters—none

10.3.2 General Parameters

TESTS	Allowed value is 1
ERROR_LIMIT	Allowed values are 1 through 32767 or "-"
PASS_LIMIT	Allowed values are 1 through 32767 or "-"
REPORT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
INHIBIT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
SCALE	Allowed values are 0 through 999
MINUTES_LIMIT	Allowed values are 0 through 32767 or "-"
ABBREVIATION	Allowed values are 0 through 4

10.4 PRESET PARAMETER VALUES FOR IOPM

TESTS = 1

ERROR_LIMIT = 5

PASS_LIMIT = -

REPORT = LOG

INHIBIT = PASSNUMBER,TESTNUMBER,SUSPENDED

SCALE = 0

MINUTES_LIMIT = -

ABBREVIATION = 4

10.5 MODE CHARACTERISTICS FOR IOPM

<u>System Mode</u>	<u>Programs Running</u>	<u>Tests Running</u>	<u>Delay Between Tests</u>
Subsystem	Only one	Only one	None
Module	One or more	Only one	Random
Exerciser	One or more	Only one	Random

10.6 INDIVIDUAL IOPM TESTS

(System Modes)

Test 01 Event Monitor

SUB, MOD

Scans the event list of IOPs configured in the System List while they are in the Idle State.

Section 11 - ANALOG INPUT HISTOGRAM PROGRAM (HIST)

11.1 HIST TEST ELEMENTS

This program evaluates one or more points on one HLAI, LLAI, or LLMux IOP and their associated FTAs that are connected to a selected PMM, APMM, HPMM. External signal sources appropriate to the FTA types are required to supply test inputs.

11.2 USE INFORMATION

The single test in this program runs with one or more points on one analog input IOP. IOP_ADDRESS must be supplied.

HIST must be run for a minimum of 100 passes in order for histogram data to be displayed, as the first 100 readings are used to establish the base values against which the remaining readings are measured.

Point values are read from PVRAW in each point's database. PVRAW values are presented in terms of "percent of range" for the HLAI, millivolts for the LLAI and LLMux millivolt ranges, and milliohms for the LLMux RTD values. This program provides the option of displaying the HLAI values in either volts or percent of range (see the HLAI_DISPLAY_OPT parameter).

11.3 SETUP PARAMETERS

11.3.1 HIST-Specific Parameters

The following operator-selected values are used by one or more of the tests in this program.

IOP_ADDRESS	File and Slot address of the HLAI, LLAI, or LLMux IOP to be tested (the IOP must appear in the configuration table). See headings 1.1.3 through 1.1.5 for information on IOP selection. This value must be filled in, a value of zero (0) will be rejected. If this IOP supports redundancy and is the "backup" IOP (attached to FTA cable B), both this IOP and its redundant partner IOP must be paired up in the system (PMEX) PARTNER list. Also, see the explanation of the parameter FTA_CABLE.
MOD_TYPE	Specifies the type of analog input IOP to be tested. Allowed values are HLAI LLAI LLMux

FTA_CABLE	Identifies which FTA cable the specified IOP is connected to. In redundant IOPs, two IOPs are connected to a common FTA by one of two cables, "A" and "B." The redundant "primary" and the nonredundant IOPs always are on cable "A." If a redundant IOP on cable "B" is specified, both partners (IOPs) must be paired up previously by use of the PMEX "PARTNER" command. Allowed values are A_Cable B_Cable
POINT_MASK	Specifies which of the points/slots on the specified IOP are to be tested. Possible point ranges vary by IOP type as follows: HLAI—points 1-16 LLAI—points 1-8 LLMux—points 1-16 on FTA #1 and points 17-32 on FTA #2 (of same type—see restrictions placed by the parameters INPUT_RANGE and HISTOGRAM_OPT) For example, the entry of 1,3-5,7-8,0 selects points/slots 1, 3, 4, 5, 7, and 8 to be tested. The terminal "0" indicates end of the list and <u>must be included</u>.
INPUT_RANGE	Specifies the full operating range of the IOP appropriate to the input signal applied and the IOP/FTA type. The allowed values and their IOP restrictions are V0_5 = 0-5 volt range for HLA I V1_5 = 1-5 volt range for HLA I MV0_100 = 0-100 millivolt range for LLA I & LLMux RTD = DINRTD (0-100 ohm) range for LLMux
DISPLAY_OPT	Specifies the extent and format of the histogram data display (see heading 11.7, HIST HISTOGRAM REPORTS). Allowed values are SUMMARY—selects Histogram Summary report TABULAR—selects Histogram Tabular report

HISTOGRAM_OPT	Specifies whether the histogram(s) to be built will be individual (a histogram per point) or composite (all points into one histogram). Up to 16 individual or one composite histogram can be built (if the specified LLMux points reside on two FTAs, two composite histograms will be built, one per FTA). The allowed values are PER_POINT COMPOSITE
ENABLE or DISABLE	Allows the operator to enable or disable specific functions associated with the execution of tests. The allowed values are DEMAND_LOG—When enabled, the histogram report(s) are displayed. DEMAND_LOG is then disabled. INIT_HISTOGRAMS—When enabled, causes the histograms to be initialized after they are displayed. When disabled, the histogram data is cumulative from display to display. DISPLAY_AT_END_OF_RUN—When enabled, causes all histograms in progress to be displayed automatically when the PASS_LIMIT is reached or a STOP HIST command is entered. No display occurs if an ABORT HIST command is entered.
PERIODIC_REPORT_PERIOD	Specifies the number of scans between periodic histogram reports. A value of zero means no periodic reports will be issued.
IN_1 .. IN_8	Floating point values in volts, millivolts, or ohms in the form "xx.xxx" for up to eight unique input signals actually applied to the points. These values must be consistent with the INPUT_RANGE specified. Up to 10 characters per value can be entered, including sign and decimal point. The program attempts to match these values with the calculated mean values in the histograms to compute the "Mean Error" (ME) for the points. If no values are provided, or the values provided deviate from the mean by 1% of full range, the program assumes no match and displays N/A (Not Available) in the appropriate places in the report.

HLAI_DISPLAY_OPT Specifies the terms in which the HLAI point values are to be displayed—volts or percent of range. The HLAI values actually are read as a percent of range, but can be converted to volts for operator convenience. Allowed values are

VOLTS

PERCENT_OF_RANGE

FREQ50_60 Identifies the input ac line frequency (used for the LLAI and LLMux IOPs). Allowed values are

SIXTY

FIFTY

11.3.2 General Parameters

TESTS	Allowed value is 1
ERROR_LIMIT	Allowed values are 1 through 32767 or "-"
PASS_LIMIT	Allowed values are 1 through 32767 or "-"
REPORT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
INHIBIT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
SCALE	Allowed values are 0 through 999
MINUTES	Allowed values are 0 through 32767 or "-"
ABBREVIATION	Allowed values are 0 through 4

11.4 PRESET PARAMETER VALUES FOR HIST

IOP_ADDRESS = 0
MOD_TYPE = HLAI
FTA_CABLE = A_CABLE
POINT_MASK = 1-16,0
INPUT_RANGE = V0_5
DISPLAY_OPT = SUMMARY
HISTOGRAM_OPT = PER_POINT
ENABLE = DISPLAY_AT_END_OF_RUN
DISABLE = DEMAND_LOG,INIT_HISTOGRAMS
PERIODIC_REPORT_PERIOD = 0
IN_1 = ?
IN_2 = ?
IN_3 = ?
IN_4 = ?
IN_5 = ?
IN_6 = ?
IN_7 = ?
IN_8 = ?
HLAI_DISPLAY_OPT = VOLTS
FREQ50_60 = SIXTY
TESTS = 1
ERROR_LIMIT = 5
PASS_LIMIT = -
REPORT = LOG
INHIBIT = PASSNUMBER,TESTNUMBER,SUSPENDED
SCALE = 0
MINUTES_LIMIT = -
ABBREVIATION = 4

11.5 MODE CHARACTERISTICS FOR HIST

System Mode	Programs Running	Tests Running	Delay Between Tests
Subsystem	Only one	Only one	None
Module	One or more	Only one	Random
Exerciser	One or more	Only one	Random

11.6 INDIVIDUAL HIST TESTS

(System Modes)

Test 01 Histogram Test

SUB, MOD, EXER

Validates unique parameters IOP_ADDRESS, MOD_TYPE, INPUT_RANGE, POINT_MASK, FTA_CABLE, and HISTOGRAM_OPT. The program alarms if there are any discrepancies, then aborts immediately.

The points specified in POINT_MASK are set up and activated. Any points that cannot be set up or that provide invalid point values are alarmed and are dropped from further testing. The program is terminated if all specified points are dropped. Any ALARM or SOFT FAIL events reported by the IOP are displayed. The required number of histograms are provided and initialized.

The first 100 readings for each histogram are averaged and become the Base value. The Base value becomes the midpoint about which the distribution of the first 100 and all subsequent readings are plotted. The Standard Deviation (SD) of the first 100 readings is used to establish the range of the pockets above and below the Base. All readings, including the first 100 are allocated to the appropriate pockets. No histogram reports are displayed until the Base has been established.

All subsequent readings are allocated to the appropriate pockets of the appropriate histograms. Histogram reports are displayed as established by ENABLE/DISABLE values and the PERIODIC_REPORT_PERIOD. The IOP event list is monitored and all events are displayed.

For the purpose of counting passes for "per point" histograms, one pass is completed when one reading is taken from all specified points. For composite histograms, one pass is counted for each reading, but the pass is not considered complete until all points have been read. Thus, for composite histograms, the pass count or the periodic report period will be a multiple of the number of points being read.

Readings are made based on individual IOP scan rates. LLAI points are scanned every 125 milliseconds, HLAI points are scanned every 250 milliseconds, and LLMux points are scanned every second.

11.7 HIST HISTOGRAM REPORTS

Following are examples of the two histogram reports available as selected by the DISPLAY_OPT parameter value.

11.7.1 Histogram Summary Report

The following is an example of a summary report for point 2 on an HLAI IOP in file 2, slot 14.

```
DD-MMM-YYYY HH:MM:SS  NMA nn  COMM nn  HIST n  HLAI  IOP 214  POINT 2
BASE  = 0.9949 volts  sd = 0.000235 volts  NUM READINGS =    100
MEAN  = 0.9949 volts  SD = 0.000233 volts  NUM READINGS =    200
INPUT = 0.9954 volts  3s = 0.003750 volts  RANGE = 0-5 volts
RE %FR = 0.0140  ME %FR = -0.0108  SKEW = -0.00002  SKEWNESS = -0.20813
LOWEST READING = 0.9945 volts  HIGHEST READING = 0.9952 volts

                        POCKET COUNTS
< SPEC < -3.0  -3.0  -2.0  -1.0  BASE  +1.0  +2.0  +3.0  > +3.0 > SPEC
-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
      0         0         0         0       75      61      64         0         0         0         0
```

In the above, BASE is the average of the first 100 readings, "sd" is the standard deviation of the first 100 readings from the Base, and NUM READINGS is the number of readings in the Base.

MEAN is the average of the first 200 readings (the Base 100 plus 100 more) and SD is the standard deviation of the 200 readings from the MEAN.

INPUT is one of the values entered in IN1-IN8, "3s" indicates the 3-sigma or maximum deviation based on an accuracy specification of .075 %FR, and the operating range is 0-5 volts.

RE %FR is the repeatability error value as a percentage of full range and ME %FR is the mean error (also known as gain error) as a percentage of full range. SKEW (same as 3rd moment) and SKEWNESS both reflect on the symmetry of distribution about the Base.

Under POCKET COUNTS are shown the number of readings for each of the pockets, which are spaced at plus/minus one, two, and three times the standard deviation from the Base. The pockets labeled "< -3.0" and "> +3.0" hold the number of readings less than/greater than three times standard deviation, up to and including the maximum deviation. The pockets labeled < SPEC and > SPEC hold the number of readings that were out of specification.

11.7.2 Histogram Tabular Report

The following is an example of a tabular report for the same point and data as in the summary report in the prior example.

```
DD-MMM-YYYY HH:MM:SS  NMA nn  COMM nn  HIST n  HLAI  IOP 214  POINT 2
BASE  = 0.9949 volts  sd = 0.000235 volts  NUM READINGS =    100
MEAN  = 0.9949 volts  SD = 0.000233 volts  NUM READINGS =    200
INPUT = 0.9954 volts  3s = 0.003750 volts  RANGE = 0-5 volts
RE %FR = 0.0140  ME %FR = -0.0108  SKEW = -0.00002  SKEWNESS = -0.20813
LOWEST READING = 0.9945 volts  HIGHEST READING = 0.9952 volts

POCKET SD RANGE      POCKET VALUE RANGE      COUNT      GRAPHIC
-----
BELOW Spec.          0.9911 - .....          0  |
-3.01 to Spec.       0.9911 to 0.9941          0  |
-3.00 to -2.51       0.9941 to 0.9943          0  |
-2.50 to -1.51       0.9943 to 0.9945          0  |
-1.50 to -0.51       0.9945 to 0.9947          75 |=====
-0.50 to +0.49       0.9947 to 0.9950          61 -|=====
+0.50 to +1.49       0.9950 to 0.9952          64 |=====
+1.50 to +2.49       0.9952 to 0.9954          0  |
+2.50 to +3.00       0.9954 to 0.9956          0  |
+3.01 to Spec        0.9956 to 0.9986          0  |
ABOVE Spec           0.9986 + .....          0  |
```

This report contains the same information as the summary report preceding but with more detail on the makeup of the pockets plus the graphic section that illustrates the relationships of the pocket counts to each other. The midpoint, or Base pocket, is the "-0.50 to +0.49" pocket and is indicated with a "-" just to the left of the graphic base line.

Section 12 - DIGITAL INPUT SEQUENCE OF EVENTS (DSOE)

12.1 DSOE TEST ELEMENTS

This program monitors Digital Input Sequence Of Events IOPs and their associated FTAs that are connected to a selected PMM, APMM, HPMM.

12.2 USE INFORMATION

DSOE will be visible to the operator only if the APMM/HPMM will be transmitting I/O Link timesynchs during the test.

Only one APMM/HPMM node, of a redundant pair, will transmit I/O Link timesynchs. If PMT3 is loaded into both APMM/HPMM nodes of a redundant pair, only one node will transmit I/O Link timesynchs until both nodes have been reset (i.e., if the node was running, DSOE stops functioning, you must reset and reload the node that stopped functioning, then run DSOE from that node or reset both nodes and reload at least one of them.

Each test will report a DSOE message of “Setup Complete — Begin Testing,” when all DISOE IOPs are ready to record events.

The tests in this program run on a single digital-input sequence of events IOP, if a value for IOP_ADDRESS is supplied, or on all configured DISOE IOPs, if that parameter value equals zero.

A pass for this test consists of processing one I/O Link transfer (up to 249 bytes) of PVCL packets, and one I/O Link transfer of ECL entries, from each of the DISOE IOPs under test for each valid test in the test list (i.e., a test will run in the current system mode/setup).

12.3 SETUP PARAMETERS

12.3.1 DISOE Specific Parameters

The following operator-selected values are used by one or more of the tests in this program.

IOP_ADDRESS	File and Slot input by operator (e.g., 201 = file 2, slot 1) of the DISOE(s) under test. The file and slot are based on files 1-8 and slots 1-15 (the IOP must appear in the configuration table). See headings 1.1.3 through 1.1.5 for information on IOP selection. A value of 0 for this parameter calls for testing of all DISOE IOPs contained in the configuration table. The default in this parameter is 0. This parameter can be changed only when DSOE is “off.”
--------------------	--

**TIME_BETWEEN
REPORTS**

Allowed values are 0 through 32767 (seconds). A value of 0 seconds means that no periodic reports will be issued. The default value for this parameter is 10. This parameter can be changed at any time.

ENABLE or DISABLE

Allows the operator to enable or disable specific functions associated with the execution of tests. The only allowed entry value is

DEMAND_LOG—when demand is enabled a report is produced as soon as possible; the program then disables demand. Initially the Demand_Log will be disabled.

This parameter can be changed at any time

POINT_MASK

The DISOE points to be tested (1-32). A terminal 0 indicates the end of the test point input. If only a terminal zero is entered, an error message will be displayed and DISOE will abort. The default value for this parameter will be 1-32,0. This parameter can be changed only when DSOE is “off.”

DEBOUNCE

The interval from a change of state until a new event is detected, during which the input to the point must remain at the new level, if it is to be reported as a change of state. The interval range is 0 to 50 milliseconds. A debounce interval of 0 results in any scanned change of state being reported as an event., The default value for this parameter is 0. This parameter can be changed only when DSOE is "off."

POINT_MODE

Allows the operator to configure the points as Status or Latched type inputs. Status type inputs report all events, after debounce, as they occur. Latched type inputs have the feature whereby a change in state, from OFF to ON, is held in the position for 1.5 seconds. The default value for this parameter is Status. This parameter can be changed only when DSOE is “off.”

PV_CHANGE DELAY

The interval (0 to 60 seconds) during which PV Change Events will be ignored after the last reported PV Change Event. The default value for this parameter is 0. This parameter can be changed only when DSOE is “off.”

PERIOD_BETWEEN EVENTS	The expected interval, in milliseconds, between events (5 to 100). The default value for this parameter will be 20 ms. This parameter can be changed only when DSOE is "off."
TOLERANCE	The acceptable tolerance, in microsecond increments, for the Peroid_Between_Events (1 to 100). The default value for this parameter will be 5. This parameter can be changed only when DSOE is "off."
COUNT	The expected number of events per point (1 to 32767, or "-"). If the input is "-", then the test will continue to evaluate events until stopped by the operator. The default value for this parameter will be 2. This parameter can be changed only when DSOE is "off."
TIMEOUT	The maximum amount of time, in seconds (1 to 32767), that an IOP can go without receiving an event and still not fail the test because of a time out error. The default value for this parameter will be 30. This parameter can be changed only when DSOE is "off."

12.3.2 General Parameters

TESTS	Allowed value is 1 through 3
ERROR_LIMIT	Allowed values are 1 through 32767 or "-"
PASS_LIMIT	Allowed values are 1 through 32767 or "-"
REPORT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
INHIBIT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
SCALE	Allowed values are 0 through 999
MINUTES_LIMIT	Allowed values are 0 through 32767 or "-"
ABBREVIATION	Allowed values are 0 through 4

12.4 PRESET PARAMETER VALUES FOR DSOE

IOP_ADDRESS = 0

TIME_BETWEEN_REPORTS = 10

ENABLE OR DISABLE = DISABLE

POINT_MASK = 1-32,0

DEBOUNCE = 0

POINT_MODE = STATUS

PV_CHANGE DELAY = 0

PERIOD_BETWEEN_EVENTS = 20 MS

TOLERANCE = 5

COUNT = 2

TIMEOUT = 30

TESTS = 1-3

ERROR_LIMIT = 5

PASS_LIMIT = -

REPORT = LOG

INHIBIT = PASSNUMBER,TESTNUMBER,SUSPENDED

SCALE = 100

MINUTES_LIMIT = -

ABBREVIATION = 4

12.5 MODE CHARACTERISTICS FOR DSOE

<u>System Mode</u>	<u>Programs Running</u>	<u>Tests Running</u>	<u>Delay Between Tests</u>
Subsystem	Only one	Only one	None
Module	One or more	Only one	Random
Exerciser	One or more	Only one	Random

12.6 INDIVIDUAL DSOE TESTS

(System Modes)

Test 01 PV Change Event Analysis

SUB, MOD

This test requires a signal source that produces a known number of events, in 32 event increments, with a known period between each pair of events. The first event is distributed to point 1 of each IOP under test, the second event will be distributed to point 2, and so on, up to the 32nd event going to point 32. If another 32 events occur, the 33rd event is distributed to point 1, the 34th event is distributed to point 2, etc.

During each pass, this test will retrieve up to one I/O Link transfer worth of events from each IOP under test. The events will be evaluated to verify that

- the events are received in the correct order by the correct points
- the event time is correct to within some operator-determined tolerance
- no extraneous events were received.

Events for all points are evaluated regardless of the Point_Mask. Each event will be counted as one of the total events for the IOP. The IOP's total events count will accumulate up to a maximum count of 4,294,967,295. Subsequent events won't be accumulated after the total events count has reached the maximum count.

This test stops running if

- the number of passes exceeds the pass limit
- the run time exceeds the minutes limit
- the operator stops, or aborts the test
- the test is completed.

Test completion occurs 300 ms after any IOP has a total number of events equal to the expected number of events (i.e., 32 times Count). Once the test is complete

- an extraneous event alarm will be displayed for any IOP with more than the expected number of events
- a test summary will be reported.

The unique parameters applicable to this test are

IOP_Address	Enable	Disable
Debounce	Point_Mode	PV_Change_Delay
Period_Between_Events	Tolerance	Count
Timeout		

Test 1 is a Utility that runs in Subsystem or Module Mode, if it is the only test selected.

If an event is received out of order (i.e., event is for point 7 when we were expecting an event for point 6), an unexpected point alarm will be displayed as follows:

DD-MMM-YYYY HH:MM:SS NMA nn COMM nn DSOE n Alarm nnn IOP Fss

PV CHANGE EVENT ANALYSIS TEST - UNEXPECTED POINT

EXPECTED - POINT: nn TIME FROM: HH:MM:SS.nnnn TO: HH:MM:SS.nnnn

RECEIVED - POINT: nn TIME:MM::SS.nnnn

If a point received is correct, but the event time is incorrect (i.e., $\text{Event_Time} < \text{Period_Between_Events} - \text{Tolerance}$, or $\text{Event_Time} > \text{Period_Between_Events} + \text{Tolerance}$), an event time error alarm will be displayed as follows:

DD-MMM-YYYY HH:MM:SS NMA nn COMM nn DSOE n Alarm nnn IOP Fss

PV CHANGE EVENT ANALYSIS TEST - EVENT TIME ERROR

EXPECTED - POINT: nn TIME FROM: HH:MM:SS.nnnn TO: HH:MM:SS.nnnn

RECEIVED - POINT: nn TIME:MM:SS.nnnn

If an IOP hasn't received an event within a timeout period, a timeout alarm will be displayed as follows:

DD-MMM-YYYY HH:MM:SS NMA nn COMM nn DSOE n Alarm nnn IOP Fss

PV CHANGE EVENT ANALYSIS TEST - IOP TIMED OUT WITHOUT AN EVENT

EXPECTED - POINT: nn TIME FROM: HH:MM:SS.nnnn TO: HH:MM:SS.nnnn

RECEIVED - POINT: nn TIME:MM:SS.nnnn

If an IOP receives an event after a time out occurred, a timeout recovery message will be displayed as follows:

PMEX MESSAGES - PV CHANGE EVENT ANALYSIS TEST

DISOE IOP; Fss recovered from time out with event - Point: Time: HH:MM:SS.nnnn

Once the test is complete, an extraneous event alarm, for each IOP with more than the expected number of events, will be displayed as follows:

DD-MMM-YYYY HH:MM:SS NMA nn COMM nn DSOE n Alarm nnn IOP Fss

PV CHANGE EVENT ANALYSIS TEST - UNEXPECTED POINT

Total events: nnnnnnnnnn

If the PV Change Event count equals 32767, the next event will result in the count rolling over to one.

The unique parameters applicable to this test are

IOP_Address	Time_Between_Reports	Point_Mask
Enable	Disable	Debounce
Point_Mode	PV_Change_Delay	

Test 3 is the exerciser test, and runs in the Subsystem or Module Mode if it is the only test selected.

PMEX MESSAGES

DISOE IOP Fss

Point	# of Events	Point	# of Events	Point	# of Events	Point	# of Events
1	nnnnn	2	nnnnn	3	nnnnn	4	nnnnn
5	nnnnn	6	nnnnn	7	nnnnn	8	nnnnn
9	nnnnn	10	nnnnn	11	nnnnn	12	nnnnn
13	nnnnn	14	nnnnn	15	nnnnn	16	nnnnn
17	nnnnn	18	nnnnn	19	nnnnn	20	nnnnn
21	nnnnn	22	nnnnn	23	nnnnn	24	nnnnn
25	nnnnn	26	nnnnn	27	nnnnn	28	nnnnn
29	nnnnn	30	nnnnn	31	nnnnn	32	nnnnn

12.7 EVENT CIRCULAR LIST REPORT

DSOE reports alarms and soft failures from the Event Circular List, for channels in the point mask, with messages such as “Data Path Check Circuit Diagnostic Failed.”

DD-MMM-YYYY HH:MM:SS NMA COMM nn DSOE n Alarm nnn IOP Fss

<Alarm comment> LOOP nnn Pass nnnn

Event Record Contents \$HH \$HH \$HH

Point had an “Alarm Event; Data Path Check Circuit Diagnostic Failed”

12.8 DSOE MESSAGES

DSOE reports operational information such as “No IOPs are available for testing.”

DD-MMM-YYYY HH:MM:SS NMA COMM nn DSOE n No IOPs are available for testing

Section 13 - SERIAL INTERFACE MODULE TESTS (SIMT)

13.1 SIMT TEST ELEMENTS

This program tests Serial Interface/Serial Device Interface input modules and their associated FTAs that are connected to selected PMM/APMMs.

13.2 USE INFORMATION

These are go no-go types of tests. At the end of each of the first eight tests, a status check is made and the pass/fail determination is made by interpreting the status. The was/should be status hexadecimal characters are displayed as a part of the error message.

Tests 9 and 10 are to be run on UDC 6000 devices. The tests should run only a minimum of times because they cause writes to the EEPROM on the UDC 6000. The limited life of this device is dependent on the number of writes to it. The writes occur once on the UDC 6000 when the RUN command is issued for SIMT (the start of each PASS does not result in a write to the UDC EEPROM).

Instructions for UDC 6000 Setup are found in subsection 13.8.

13.3 SETUP PARAMETERS

13.3.1 SIMT-Specific Parameters

The following operator-selected values are used by one or more of the tests in this program.

IOP_ADDRESS File and Slot input by operator (e.g., 201 = file 2, slot 1), of the SIMT(s) under test. The file and slot are based on files 1-8 and slots 1-15 (the IOP must appear in the configuration table). See headings 1.1.3 through 1.1.5 for information on IOP selection. A value of 0 for this parameter calls for testing of all SIMT IOPs contained in the configuration table. The default in this parameter is 0. This parameter can be changed only when SIMT is “off.”

ENABLE or DISABLE Allows the operator to enable or disable specific functions associated with the execution of tests. The only allowed functions are REPORT_IOPS and DEMAND_LOG:

REPORT_IOPS—when REPORT_IOPS is enabled, a report is produced as soon as possible; the program then disables REPORT_IOPS. Initially the REPORT_IOPS will be disabled. The report shows all IOPs and FTAs under test. This parameter can be changed at any time.

DEMAND_LOG—when DEMAND_LOG is enabled, Test 10 will log all selected PVs in Percent of Full Scale as soon as possible; the program then disables DEMAND_LOG. Initially, the

DEMAND_LOG will be enabled. The report shows all IOPs and FTAs under test. This parameter can be changed at any time.

FTA Allows the operator to select the FTAs to test. The allowed values are one, two, or both.

13.3.2 General Parameters

TESTS	Allowed value is 1 through 10
ERROR_LIMIT	Allowed values are 1 through 32767 or "-"
PASS_LIMIT	Allowed values are 1 through 32767 or "-"
REPORT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
INHIBIT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
SCALE	Allowed values are 0 through 999
MINUTES_LIMIT	Allowed values are 0 through 32767 or "-"
ABBREVIATION	Allowed values are 0 through 4

13.4 PRESET PARAMETER VALUES FOR SIMT

IOP_ADDRESS = 0

ENABLE = DEMAND_LOG

DISABLE = REPORT_IOPS

FTA = ONE

TESTS = 1-10

ERROR_LIMIT = 5

PASS_LIMIT = -

REPORT = LOG

INHIBIT = PASSNUMBER,TESTNUMBER,SUSPENDED

SCALE = 100

MINUTES_LIMIT = -

ABBREVIATION = 4

13.5 MODE CHARACTERISTICS FOR SIMT

<u>System Mode</u>	<u>Programs Running</u>	<u>Tests Running</u>	<u>Delay Between Tests</u>
Subsystem	Only one	One or more	None
Module	One or more	One or more	Random
Exerciser	One or more	Only one	Random

13.6 INDIVIDUAL SIMT TESTS

(System Modes)

Test 01 RS-232/485 Signal Loopback Test SUB, MOD

This test requires a loopback fixture, 51402236, which interconnects the Transmit and Receive data and Control lines.

485 TX Data —> Rx Data RTS —> CTS DTR —> DSR
232 TX Data—> SPARE —> DCD

Test patterns are then transmitted on these lines.

Test 02 RS-232 Fixed Data Loopback Test SUB, MOD, EXER

This test requires a loopback fixture, 51402236, which interconnects the data lines.

The test transmits a fixed data pattern via the RS-232 TX line that is then received on the RS-232 RX line. The data stream is provided by the firmware and each byte has 10 bits with 1 start bit, 8 data bits, and 1 stop bit. The data is collected and compared to the transmitted data for integrity and the status is reported. The data is transferred at a 19200 baud data rate.

Test 03 RS-232 Programmable Data Loopback Test SUB, MOD

This test requires a loopback fixture, 51402236, which interconnects the transmit and receive data lines.

The test transmits a data pattern via the RS-232 TX line that is then received on the RS-232 RX line. The data stream is provided by PMTS and each byte has 10 bits with 1 start bit, 8 data bits, and one stop bit. The data is collected and compared to the transmitted data for integrity and the status is reported. The data is transferred at a 19200 baud data rate.

Test 04 RS-485 Fixed Data Internal Loopback Test SUB, MOD

This test requires no loopback fixture; however, the loopback test fixture utilized in tests 1, 2, 3, 6, and 7 does not affect the outcome of this test and need not be removed.

The RS-485 UART internally connects the transmit and receive data lines. The test transmits a fixed data pattern via the RS-232 TX line that is then received on the RS-232 RX line. The data stream is provided by the firmware and each byte has 10 bits with 1 start bit, 8 data bits, and 1 stop bit. The data is collected and compared to the transmitted data for integrity and the status is reported. The data is transferred at a 19200 baud data rate.

Test 05 RS-485 Programmable Data Internal Loopback Test SUB, MOD

This test requires no loopback fixture; however, the loopback test fixture utilized in tests 1, 2, 3, 6, and 7 does not affect the outcome of this test and need not be removed.

The RS-485 UART internally connects the transmit and receive data lines. The test transmits a data pattern via the RS-232 TX line that is then received on the RS-232 RX line. The data stream is provided by PMTS and each byte has 10 bits with 1 start bit, 8 data bits, and 1 stop bit. The data is collected and compared to the transmitted data for integrity and the status is reported. The data is transferred at a 19200 baud data rate.

Test 06 RS-485/232 Fixed Data Loopback Test SUB, MOD

This test requires a loopback fixture, 51402236, which interconnects the 485 TX data line to the 232 RX data line.

The test transmits a fixed data pattern via the RS-485 TX line that is then received on the RS-232 RX line. The data stream is provided by the firmware and each byte has 10 bits with 1 start bit, 8 data bits, and 1 stop bit. The data is collected and compared to the transmitted data for integrity and the status is reported. The data is transferred at a 19200 baud data rate.

Test 07 RS-485/232 Programmable Data Loopback Test SUB, MOD

This test requires a loopback fixture, 51402236, which interconnects the 485 TX data line to the 232 RX data line.

The test transmits a fixed data pattern via the RS-485 TX line that is then received on the RS-232 RX line. The data stream is provided by PMTS and each byte has 10 bits with 1 start bit, 8 data bits, and 1 stop bit. The data is collected and compared to the transmitted data for integrity and the status is reported. The data is transferred at a 19200 baud data rate.

Test 08 Watch Dog Timer Test SUB, MOD

For this test, the FTA stops functioning by going into a tight infinite loop. Inside the loop, it stops refreshing the hardware based on Watch Dog Timer (WDT), but maintains communication to the test program, as well as continuing to clear the IOP to FTA Change Buffer. Eventually the WDT times out, which halts the FTA microprocessor. This test is considered to be a success if the FTA fails to maintain communications with the test program (signifying timing out, i.e., correct operation of the WDT). In the case where the FTA continues to function (communications to the test program is maintained) without the WDT latch being refreshed, this test is considered to be a failure. No separate fixture is required for this test.

NOTE

This is a utility test and must be the only test in the test list to run.

Test 09 UDC Status Test SUB, MOD

This test is a UTILITY Test. It checks the status of the UDC 6000/FTA communication.

This test will verify that the FTA type is a UDC 6000. The check will be done automatically as this test is for UDC 6000 FTAs only. If the parameter STISWVER does not contain UDC 6000, an appropriate error message will be generated and the test will be discontinued on the associated SDI IOP.

This test will check the STATE parameter for a status of OK. If the status is not OK, an error message will be displayed indicating the current STATE.

NOTE

This is a utility test and must be the only test in the test list to run.

Test 10 Analog Input Test

SUB, MOD

This test is a UTILITY Test.

This test is a limited test of the UDC 6000. It will allow analog input values to be displayed when DEMAND_LOG is enabled.

This test will verify that the FTA type is a UDC 6000. The check will be done automatically as this test is for UDC 6000 FTAs only. If the parameter STISWVER does not contain UDC 6000, an appropriate error message will be generated and the test will be discontinued on the associated SDI IOP. During configuration, this test will write the IOP Address, FTA number, and point to each point's SFITAG. This can be displayed at the UDC 6000 via the front panel control. In doing this, the operator can determine where each point is assigned to the UDC 6000s.

The test will then configure all points to analog input points with no trip point values. As in Test 09, the STATE of each point will be checked after the points have been ACTIVATED. This STATE check will produce an error if the STATE for that point is not OK. This STATE check is performed only one time and the test then starts looking for the operator to enable the DEMAND_LOG. If a log is desired, the operator needs to enable DEMAND_LOG via the ENABLE/DISABLE user-settable parameter.

NOTE

This is a utility test and must be the only test in the test list to run.

13.7 SIMT MESSAGES

SIMT reports failures in the following format:

TESTS 1 - 8

DD-MMM-YYYY HH:MM:SS NMA 4 COMM nn SIMT n Alarm ### IOP ###

Test never started on FTA FTA# n PASS n

ALL TESTS

DD-MMM-YYYY HH:MM:SS NMA 4 COMM nn SIMT n Alarm ### IOP ###

Status words are incorrect! FTA# n PASS n

WASXXXX

SB YYYY

TEST 02 through 07

DD-MMM-YYYY HH:MM:SS NMA 4 COMM nn SIMT n Alarm ### IOP ###

Reading FTSTx_STAT FTA# n PASS n

Timeout Error indicated by status incorrect.

TEST 8

DD-MMM-YYYY HH:MM:SS NMA 4 COMM nn SIMT n Alarm ### IOP ###

WDT Test Failed. FTA# n PASS n

TEST 9

DD-MMM-YYYY HH:MM:SS NMA 4 COMM nn SIMT n Alarm ##### IOP ###

Checking STATE of UDC Communications! FTA# n PASS n

SFI STATE NOT OK!

POINT # XX

STATE WAS; \$XX

STATE SB; \$XX

TEST 10

DD-MMM-YYYY HH:MM:SS NMA 4 COMM nn SIMT n Alarm ##### IOP ###

Checking STATE of UDC Communications! FTA# n PASS n

SFI STATE NOT OK!

POINT # XX

STATE WAS; \$XX

STATE SB; \$XX

(Also produces DEMAND_LOG)

NOTE: The first digit(s) of the alarm number indicates which test failed.

13.8 UDC 6000 SETUP

It will help if you have a UDC 6000 Process Controller Product Manual, but it is not required.

The firmware in the UDC 6000 must be the option that allows communication to a PM/APM system.

To do a *FULL* SIMT hardware verification test requires four UDC 6000 controllers wired per Figure 13-1 and 13-2. The Setup instructions below assume that four UDC 6000 controllers are to be tested.

Turn power on to the (up to 4) UDC 6000 controllers. Press ALM/ACK on each unit to stop the bar graph flashing.

If setup is required, or if you would like to verify it, start with controller number 1 and perform the following steps. If setup has previously been performed, the following steps are not required.

1. Press SET/UP 8 times and you will see INPUT 1 on the display.
2. Press FUNC and display XMITTER 1. Press the up arrow until you see LINEAR displayed above XMITTER 1.
3. Press SET/UP and you will see INPUT 2 on the display.
4. Press FUNC and display XMITTER 2. Press the up arrow until you see LINEAR displayed above XMITTER 2.
5. Press SET/UP and you will see INPUT 3 on the display.
6. Press FUNC and display XMITTER 3. Press the up arrow until you see LINEAR displayed above XMITTER 3.
7. Press SET/UP and you will see INPUT 4 on the display.
8. Press FUNC and display XMITTER 4. Press the up arrow until you see LINEAR displayed above XMITTER 4.
9. Press SET/UP and you will see CONTROL on the display.
10. Press FUNC and you will see PV SOURCE on the display. Press the up arrow until you see INPUT 1.
11. Press SET/UP and you will see CONTROL 2 on the display.
12. Press FUNC and you will see PV 2 SRC on the display. Press the up arrow until you see INPUT 2.
13. Press SET/UP 2 times and you will see COMMUN on the display.
14. Press FUNC and use the up arrow until you see RS-485 TDC.
15. Press FUNC and you will see SP1 ADDR. Press the up or down arrow until you see 1 in the display.

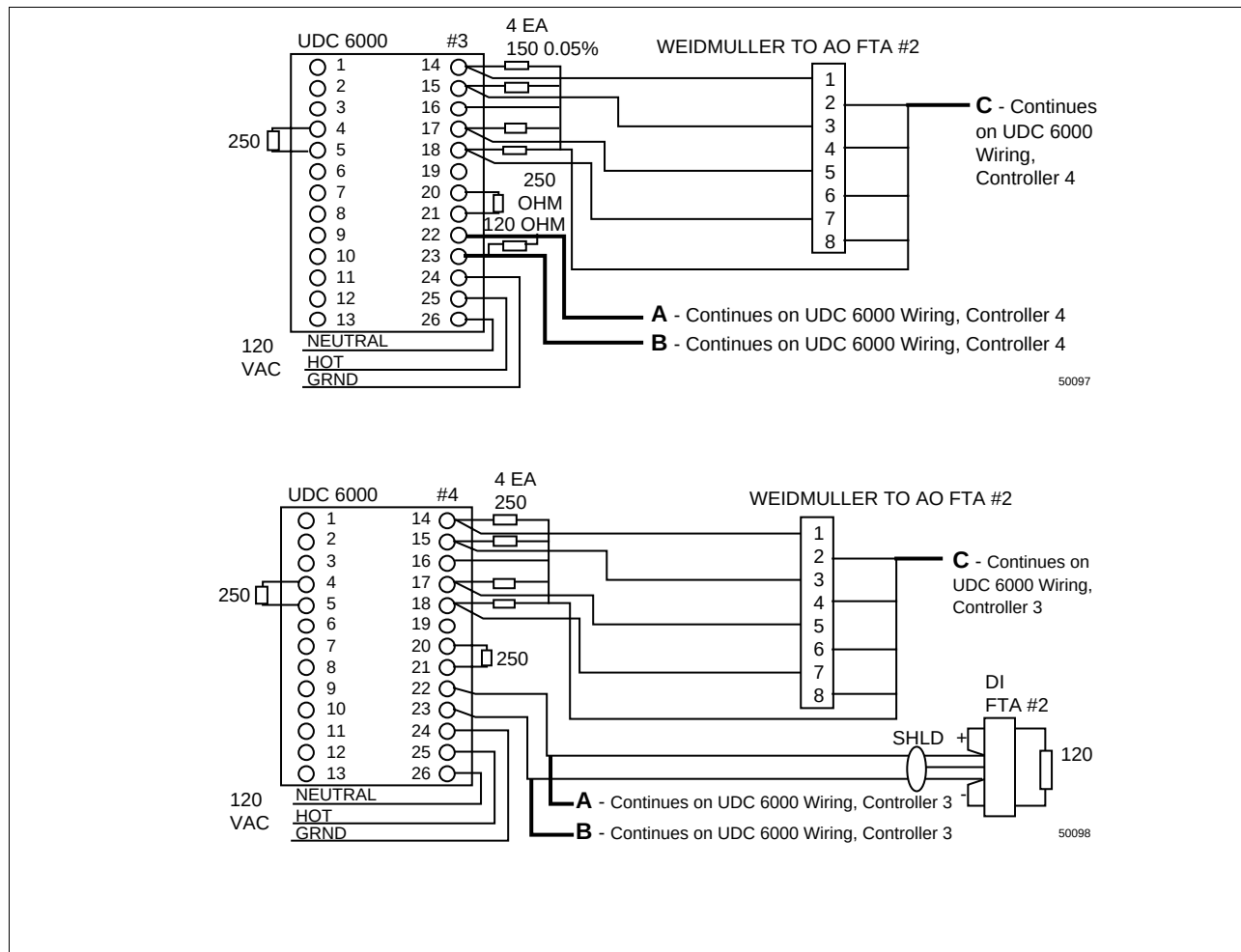
16. Press FUNC and you will see DC1 ADDR. Press the up or down arrow until you see 2 in the display.
17. Press FUNC and you will see SP2 ADDR. Press the up or down arrow until you see 3 in the display.
18. Press FUNC and you will see DC2 ADDR. Press the up or down arrow until you see 4 in the display.
19. Press LOWR/DIS to complete the setup for unit 1.
20. Repeat steps 1 to 19 for controller number 2. When you get to step 15 use the arrow key to see 5 in the display. When you get to step 16 use the arrow key to see 6 in the display. When you get to step 17 use the arrow key to see 7 in the display. When you get to step 18 use the arrow key to see 8 in the display.
21. Repeat steps 1 to 19 for controller number 3. When you get to step 15 use the arrow key to see 1 in the display. When you get to step 16 use the arrow key to see 2 in the display. When you get to step 17 use the arrow key to see 3 in the display. When you get to step 18 use the arrow key to see 4 in the display.
22. Repeat steps 1 to 19 for controller number 2. When you get to step 15 use the arrow key to see 5 in the display. When you get to step 16 use the arrow key to see 6 in the display. When you get to step 17 use the arrow key to see 7 in the display. When you get to step 18 use the arrow key to see 8 in the display.

13.8.1 Operational Description

The first two UDC 6000 controllers will be used to test SI FTA number 1. The next two controllers will be used to test SI FTA number 2.

1. Connect communication line from controllers 1 and 2 to SI FTA 1. Connect communication line from controllers 3 and 4 to SI FTA 2.
2. You are now set to run SIMT using the ATPS system by connecting AO #1 to the HLAI connectors on controllers 1 and 2, and connecting AO #2 to the HLAI connectors on controllers 3 and 4.
3. If you are using the UDC 6000 controllers in Integration, you can use any two AOs in the CVAS system as inputs to the controllers.

Figure 13-2 — UDC 6000 Wiring, Controllers 3 and 4



Section 14 - FIELDBUS TESTS (FBUS)

14.1 FBUS TEST ELEMENTS

This program tests the Fieldbus IOP (and its associated FTAs) that is connected to a selected High-Performance Process Manager HPM). A Process Manager (PM) or Advanced Process Manager (APM) will not support the Fieldbus IOP and FTA.

14.2 USE INFORMATION

The tests in this program run on a single analog Fieldbus IOP, if a value for IOP_ADDRESS is supplied, or on all configured Fieldbus IOPs, if the parameter value equals zero.

A pass for these tests consists of verifying that the IOP is executing the background diagnostics and that IOP Segment 1 can communicate with Segment 2 through a Loopback cable, and Segment 3 can communicate with Segment 4 through a Loopback cable.

Testing of redundant IOPs is not supported.

14.3 SETUP PARAMETERS

14.3.1 FBUS-Specific Parameters

The following operator-selected values are used by one or more of the tests in this program.

IOP_ADDRESS	File and Slot address of a Fieldbus IOP to be tested (the IOP must appear in the configuration table). See subsections 1.1.3 through 1.1.5 for information on IOP addressing and configuration. A value of 000 for this parameter calls for the testing of all Fieldbus IOPs contained in the configuration table.
SECM_CONFIGURE	A flag that is used during test initialization to configure the IOP. If this flag is set to YES, then the test will attempt to configure the Segment Entity Control Module (SECM) for all four Segments. This step must be performed only once after power is applied to the IOP. Test errors will result if an attempt is made to configure the SECMs that were configured during a previous test execution. If this flag is set to NO, then the test will not attempt to configure the SECMs.

LIVE_WAIT The number of seconds that the Segment Loopback test will wait between configuring the last SECM of the last IOP under test and the beginning of the request for Live List information. The allowed value is sub-ranged to be less than or equal to 240 seconds.

14.3.2 General Parameters

TESTS	Allowed values are 1 through 2
ERROR_LIMIT	Allowed values are 1 through 32767 or "-"
PASS_LIMIT	Allowed values are 1 through 32767 or "-"
REPORT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
INHIBIT	Allowed values are PASSNUMBER, TESTNUMBER, SUSPENDED, LOG
SCALE	Allowed values are 0 through 999
MINUTES_LIMIT	Allowed values are 0 through 32767 or "-"
ABBREVIATION	Allowed values are 0 through 4

14.4 PRESET PARAMETER VALUES FOR FBUS

IOP_ADDRESS = 0

SECM_CONFIGURE = YES

LIVE_WAIT = 30

TESTS = 1-2

ERROR_LIMIT = 5

PASS_LIMIT = -

REPORT = LOG

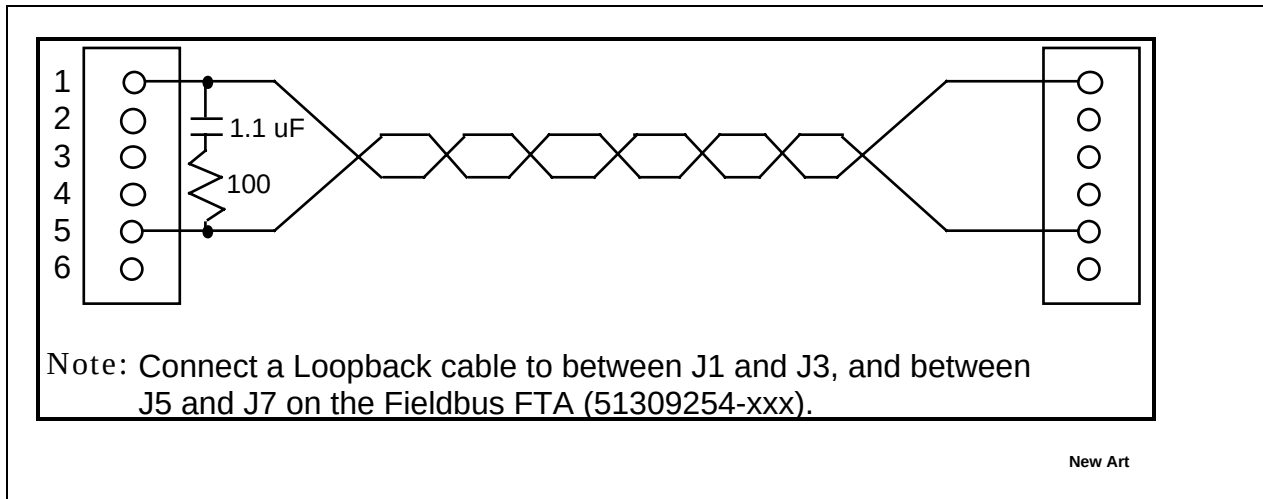
INHIBIT = PASSNUMBER,TESTNUMBER,SUSPENDED

SCALE = 0

MINUTES_LIMIT = -

ABBREVIATION = 4

Figure 14-1 — Loopback Cable



Appendix A - HARDWARE INTERACTIONS AND ALARM CODES

A.1 COMMON EVENTS AND ALARM CODE CONVENTIONS

A.1.1 Hardware Interactions of Common Events

READ SLOT PARAMETER COMMAND—When a test in the PM reads a parameter from an IOP it sends a request over the I/O link to the IOP's Device Physical Address (DPA). The device should respond with an Overall Status, the byte count of the reply, a Message Status, the parameter data type, and the actual data of the parameter. Slot 0 parameters apply to the entire card, slot 1 and above parameters apply to the actual I/O slots. For example, the parameter PVRAW for slot 4 is the raw process value (PV) for slot 4.

WRITE SLOT PARAMETER COMMAND—When a test in the PM writes a parameter to an IOP it sends the request over the I/O link to that IOP's DPA. The initial reply from the IOP contains an Overall Status and a message number. This indicates only whether the data was correctly received. To establish that the parameter write actually worked, the message number is used in a "Get Write Message Status" poll to the IOP. If the IOP responds with "In Progress," the poll is repeated until a status is returned that indicates success or some sort of failure. These types of failure will produce an error message similar to the following:

```
31-MAY-1988 00:46:47 NMA -- MM 27 COMM LLAI 2 Alarm 303 IOP 113
Configuring IOPs      LOOP 1      PASS 1
Bad Write Msg Status: WM Status = $61
Command Code $0A To Parameter Code 47 Of Slot 7
```

In this example, "Configuring IOPs" tells you what the test was trying to do at the time, "\$61" is the IOL message status (in hexadecimal), "113" is the IOP address, "0A" is the Command Code (WRITE SLOT PARAMETER in this case), "47" is the decimal number of the parameter value that was written, and "7" is the slot (or point) number. Failures of this type are not expected, and you should contact Diagnostic Software (customers contact TAC) if you see them.

When the Write Parameter command is successful, it can have further hardware interactions dependant on IOP type and which parameter was changed. These interactions can include changing the board mode, changing a slot input value, or enabling a feature of the board.

GET EVENT (or CHANGE) LIST STATUS—This request is sent to an IOP's slot 0 to see if there are Events (or Changes) in its Event (Change) circular list. The reply contains an overall status, a message status, and the list status. It should have no more hardware interaction than a READ SLOT PARAMETER.

GET EVENT (CHANGE) CIRCULAR LIST—This requests the actual list from the IOP (in terms of hardware interaction, this is much like a READ SLOT PARAMETER) and then updating its Circular List Remove Pointer (similar to WRITE SLOT PARAMETER).

EVENT SCANNING—During all tests, the Event Circular List Status is checked periodically. If events are present, they are fetched and, if allowed by the test program event mask, are reported. The test program event mask provides three choices of event displays: all events except system events, all events except bracket events, or all events. The mask used depends on what test is running and sometimes on what the test is doing at the time.

Two examples of event displays follow:

```
31-MAY-1988 00:04:30 NMA -- MM 3 COMM LLAI 2 Alarm 204 IOP 113
Configuring IOPs      LOOP 1      PASS 1
Event Record contents = $17 $87 $1D
Point 7 had a "Soft Failure" Event A/D CONVERSION TIMED OUT
```

```
31-MAY-1988 00:14:01 NMA -- MM 3 COMM DIMT 2 Alarm 140 IOP 115
Reading Process Values LOOP 1      PASS 199
Event Record contents = $17 $80 $06
Module had a "Soft Failure" Event FTA MISSING
```

Events recorded in the Event Circular List include

- System Events—An example is module state going to RUN or IDLE
- Bracket Events—An example is PTEXECST going ACTIVE or INACTIVE
- Soft Failure Events—Examples are FTA not present, Bad PV, etc.
- Process Alarm Events—Examples are high/low level trip points

A.1.2 Alarm Codes Conventions

In PMTS tests, the alarm text indicates what the test program was doing during the failure, what went wrong, and pertinent status information. The Alarm Codes generally ascend as we get further into a test. If a software "bug" is suspected, this information can be valuable in tracking down the area of test software involved. Alarm Codes are usually less helpful than the message in diagnosing a hardware failure.

For all PMTS tests, the first one or two digits are the test number; the next two digits indicate the area of the test that failed. Examples:

Alarm Code 1210 is a failure detected in test 12.

Alarm Code 105 is a failure detected in test 1.

A.2 AOMT INTERACTIONS AND ALARM CODES

A.2.1 Hardware Interactions of AOMT

INITIALIZATION—Each time the RUN command is entered, AOMT performs the following before any tests are run:

- A. Checks the module type and status bytes (using READ SLOT PARAMETER) of each AO IOP to be tested. If the module type is wrong, or if the module status is bad, the reason(s) is reported and the Event Circular List is read and displayed, if possible. If the module type and status are both correct, the IOP is included in a list of AO IOPs to be tested.
- B. Configures each AO IOP to be tested. This procedure consists of using WRITE SLOT PARAMETER to put all slots (slot = 0) in each module into a HOLD FailOpt condition in case a failure occurs; then, a WRITE SLOT PARAMETER is used to change all modules from IDLE to RUN state. A WRITE SLOT PARAMETER is sent to each module point (slot) to deactivate the slot, give the desired direction of the output curve slope, and then reactivate the slot. Then, a WRITE SLOT PARAMETER is sent to each point (slot) on each testable module to put the slot's output current at its low-limit value (-6.9%).
- C. If everything has been successful, all events in the Event Circular List are gathered and discarded because the events and changes created during configuration can be ignored.

If there are any I/O link failures during configuration or testing of an AO IOP, the IOP is removed from the list of AO IOPs to be tested and that action is reported.

TEST 1—During each pass a WRITE SLOT PARAMETER is sent repeatedly to each slot of all testable modules to write a value to each output. After each write process (maximum of eight writes), the module status and Event Circular List are checked (see EVENT SCANNING). All the events are reported. The IOP is removed from the AO IOP test list if status or Event Circular List errors occur.

TEST 2—Once during each pass of this test, a WRITE SLOT PARAMETER is sent to each slot of a randomly selected AO module to write a randomly selected value to its outputs. After each write process (maximum of eight writes), the module status and Event Circular List are checked (see EVENT SCANNING). All the events are reported. The IOP is removed from the AO IOP test list if status or Event Circular List errors occur.

TEST 3—Once during each pass of this test, a WRITE SLOT PARAMETER is sent to each slot of all testable AO modules to write an operator-specified value to the outputs. After each write process (maximum of eight writes), the module status and Event Circular List are checked (see EVENT SCANNING). All the events are reported. The IOP is removed from the AO IOP test list if status or Event Circular List errors occur.

A.2.2 AOMT Alarm Codes

INITIALIZATION

- 20-21 Failures related to the check of IOP type of a single operator-specified AO module during building of the AO testable module list.
- 22-28 Failures related to status check of a single operator-specified AO module during building of the AO testable module list.
- 29-34 Failures related to clearing the event circular list of a single operator-specified AO module during building of the AO testable module list.
- 35-49 Failures related to display of the event circular list of a single operator-specified AO module during building of the AO testable module list.
- 50-51 Failures related to check of the IOP type of AO modules from the system IOP list during the building of the AO testable module list.
- 52-58 Failures related to status check of AO modules from the system IOP list during the building of the AO testable module list.
- 59-64 Failures related to clearing the event circular list of AO modules from the system IOP list during the building of the AO testable module list.
- 65-74 Failures related to display of the event circular list of AO modules from the system IOP list during the building of the AO testable module list.
- 75-81 Failures related to write-slot-parameter to the AO IOPs listed in the AO testable list to put the AOs in a HOLD, RUN, INACTIVE, DIRECT-polarity, REVERSE-polarity, ACTIVE and/or low-current value configuration.
- 82-87 Failures related to clearing the event circular list of AO modules during the configuration of the AO modules on the AO testable module list.
- 88-99 Failures related to display of the event circular list of AO modules during the configuration of the AO modules on the AO testable module list.

TEST 01—OUTPUT CHECK

- 110-119 Failures related to write-real-parameter output values to AO module slots.
- 120-129 Failures related to checking status of AO modules.
- 130-139 Failures related to clearing the event circular list.
- 140-149 Failures related to display of the event circular list.

TEST 02—RANDOM OUTPUT

210-219 Failures related to write-real-parameter output values to AO module slots.

220-229 Failures related to checking status of AO modules.

230-239 Failures related to clearing the event circular list.

240-249 Failures related to display of the event circular list.

TEST 03—ACCURACY CHECK

310-319 Failures related to write-real-parameter output values to AO module slots.

320-329 Failures related to checking status of AO modules.

330-339 Failures related to clearing the event circular list.

340-349 Failures related to display of the event circular list.

A.3 DIMT INTERACTIONS AND ALARM CODES

A.3.1 Hardware Interactions of DIMT

NOTE

Because of memory constraints, DIMT and PIMT tests cannot run simultaneously.

INITIALIZATION—Each time the RUN command is entered, DIMT performs the following before any tests are run:

- A. Checks the module type and status bytes (using READ SLOT PARAMETER) of each DI IOP to be tested. If the IOP type is wrong or the IOP status is bad, the reason(s) is reported and the Event Circular List is read and displayed, if possible. If the IOP type and status are both correct, the IOP is included in a list of DI IOPs to be tested.
- B. Configures each DI IOP to be tested. WRITE SLOT PARAMETER is used to deactivate all slots, turn on change reporting, and reactivate the slots. Then, a WRITE SLOT PARAMETER to slot 0 puts the module in RUN mode.
- C. All events in the Event Circular List and all changes in the Change List are gathered and discarded because the events and changes created during configuration can be ignored.

If there are any I/O link failures during configuration or testing of an IOP, the IOP is removed from the list of IOPs to be tested and that action is reported.

TEST 1—During each pass, the Event Circular List and the Change Circular List are checked repeatedly (see EVENT SCANNING). All events are reported (the mask for "all events" is used). Changes from the Change Circular List are used to update the change counts for the Count Report. The Change List is checked again to obtain any further changes.

TEST 2—During each pass the Event Circular List is checked repeatedly (see EVENT SCANNING). All events are reported (the mask for "all events" is used). Once per second, the parameter PVAL is fetched with a Read Slot Parameter. The fetch is repeated until there are no further changes. PVAL is used in the VALUE REPORT because it contains the process values for all 32 slots.

A.3.2 DIMT Alarm Codes

The final two digits of the alarm indicate the area where the failure was detected as follows:

00-49 Failures in the test programs or subroutines (numbers divisible by 5 are in a test or an immediate subroutine, others are from deeper in the subroutines).

50-69 Failures detected during initialization.

70-74 Failures while building the IOP list (and checking those IOPs for correct type and status).

75-79 Failures while configuring the cards and cleaning out event lists.

80-89 Failures while collecting and trashing Change Lists.

90-99 Failures in module abort and cleanup.

A.4 DOMT INTERACTIONS AND ALARM CODES

A.4.1 Hardware Interactions of DOMT

INITIALIZATION—Each time the RUN command is entered, DOMT performs the following before any tests are run:

- A. Checks the module type and status bytes (using READ SLOT PARAMETER) of each DO IOP to be tested. If the module type is wrong or if the module status is bad, the reason(s) is reported and the Event Circular List is read and displayed, if possible. If the module type and status are both correct, the IOP is included in a list of DO IOPs to be tested.
- B. Configures each DO IOP to be tested. This procedure consists of using WRITE SLOT PARAMETER to deactivate all slots, set the IOP (slot 0) to RUN, and set the DO type for each slot to either Pulse Width Modulated or Status, depending on whether parameter DO_MODE is set to PULSED or LATCHED/MOMENTARY, respectively (DO_MODE is assigned LATCHED for Tests 1 and 2). Finally, all slots are activated.
- C. All events in the Event Circular List are read and all alarm events are displayed. Bracket and system events are discarded because these are created during configuration and can be ignored.

If there are any I/O link failures during configuration or testing of an DO IOP, the IOP is removed from the list of DO IOPs to be tested and that action is reported.

TEST 1—During each pass

- A. Module status is checked for communications errors or Hard Failures. If communications are encountered, the RESET COMMUNICATION ERRORS command is broadcast to all IOPs on the I/O link.
- B. The Event Circular List is checked and all events are displayed (see EVENT SCANNING).
- C. Using the WRITE SLOT PARAMETER command, the DO points/slots are set to their appropriate state depending on the selected PATTERN. If PATTERN is either USER_DEFINED or WIG_WAG, all points on the DO IOP are set to their appropriate state before moving to the next DO IOP; for all other PATTERN selections, the same point on all DO IOPs is completed before the next higher point is affected (i.e., point 1 on all DO IOPs is set, then point 2 on all DO IOPs, etc.).
- D. If any I/O link errors occur, the faulty DO IOP is removed from the local list of DO IOPs to be tested. The completion of the pattern on all DO IOPs constitutes one pass of the test.

TEST 2—During each pass, the same operations as in Test 1 are performed on one DO IOP selected at random from the list of available DO IOPs.

TEST 3—During each pass, the same operations as in Test 1 are performed on all available DO IOPs. The difference is that, in Test 3, points can be LATCHED, PULSED or MOMENTARY (see parameter DO_MODE), and POINT_MASK is effective.

A.4.2 DOMT Alarm Codes

The final two digits of the alarm indicate the area where the failure was detected as follows:

01	Module not responding as a DOM.
02-04	I/O link transfer error during check of DOM status.
05	DOM state not idle or run.
06	FTA not present.
07	DOM has soft failure.
08	DOM has hard failure.
09	DOM has comm failure.
10	I/O link transfer error while setting points inactive.
11-13	I/O link transfer error while clearing the event list.
14	I/O link transfer error during read/display of event list.
50	I/O link transfer error while setting points active.
51	I/O link transfer error while setting hold mode.
52	I/O link transfer error while setting module to run mode.
53	I/O link transfer error while setting points inactive.
54	I/O link transfer error while setting DOTYPE to PWM.
55	I/O link transfer error while setting point's pulse period.
56	I/O link transfer error while setting DOTYPE to STATUS.
57	I/O link transfer error while setting points active.
60-62	I/O link transfer error while checking event list.
64	I/O link transfer error during read/display event list.
66	I/O link communications error detected by module.
70	I/O link transfer error while setting points to OFF state.
71	I/O link transfer error while setting points to ON state.
72	I/O link transfer error while writing to ONPULSE parameter.
73	I/O link transfer error while writing to OP parameter.
74	I/O link transfer error while setting all points to OFF in pulse mode.
75	I/O link transfer error while setting all points to OFF at end of test.

A.5 HLAI INTERACTIONS AND ALARM CODES

A.5.1 Hardware Interactions of HLAI

INITIALIZATION—Each time the RUN command is entered, HLAI performs the following before any tests are run:

- A. Checks the module type and status bytes (using READ SLOT PARAMETER) of each HLAI IOP to be tested. If the module type is wrong or if the module status is bad, the reason(s) is reported and the Event Circular List is read and displayed, if possible. If the module type and status are both correct, the IOP is included in a list of HLAI IOPs to be tested.
- B. Configures each HLAI IOP to be tested. This procedure consists of using WRITE SLOT PARAMETER to each slot to deactivate all slots, set the engineering span, set the sensor type, set the characterization, set the alarm deadband to 0.5 percent, and set the required trip points; then, a WRITE SLOT PARAMETER to slot 0 puts the module in RUN mode.
- C. All events in the Event Circular List are gathered and discarded because the events and changes created during configuration can be ignored.
- D. Reactivates the slots by a WRITE SLOT PARAMETER to PTEXECST.
- E. Gathers and displays all Alarm Events in the event list.

If there are any I/O link failures during configuration or testing of an HLAI IOP, the IOP is removed from the list of HLAI IOPs to be tested and that action is reported.

TEST 1—During each pass the Event Circular List is checked (see EVENT SCANNING) and all events are reported. If DEMAND_LOG is enabled, process values (PVCALC) are read for each point and displayed in a log in UNIT_OF_MEASURE terms. DEMAND_LOG then is disabled.

TEST 2—During each pass the Event Circular List is checked (see EVENT SCANNING).

A.5.2 HLAI Alarm Codes

The final two digits of the alarm indicate the area where the failure was detected as follows:

- | | |
|-------|---|
| 00-30 | Failures while trying to build the IOP list used by the test. |
| 31-89 | Failures during configuration of the IOPs. |
| 90-99 | Failures while scanning the event list. |

A.6 LLAI INTERACTIONS AND ALARM CODES

A.6.1 Hardware Interactions of LLAI

INITIALIZATION—Each time the RUN command is entered, LLAI performs the following before any tests are run:

- A. Checks the module type and status bytes (using READ SLOT PARAMETER) of each LLAI IOP to be tested. If the module type is wrong or if the module status is bad, the reason(s) is reported and the Event Circular List is read and displayed, if possible. If the module type and status are both correct, the IOP is included in a list of LLAI IOPs to be tested.
- B. Configures each LLAI IOP to be tested. This procedure consists of using WRITE SLOT PARAMETER to each slot to deactivate all slots, set the engineering span, set the sensor type, set the characterization, set the alarm deadband (PVALDB) to 0 percent, and set the required trip points. If the sensor type is thermocouple, the extended option is set and, unless FILTER_DELAY is not "?" or 0.0, TF is set to the FILTER_DELAY value. If the sensor type is 0-100 MV, PVRAWHI and PVRAWLO are set, a WRITE SLOT PARAMETER to slot 0 puts the module in RUN mode.
- C. All events in the Event Circular List are gathered and discarded because the events and changes created during configuration can be ignored.
- D. Reactivates the slots by a WRITE SLOT PARAMETER to PTEXECST.
- E. Gathers and displays all Alarm Events in the event list.

Note that it is possible to get "soft fail" recovered events when INPUT_RANGE = AUTO. The points are first configured to MV0-100 sensor type and if a point has a high voltage applied to it, a "soft fail" occurs. The test then reconfigures the point to V0_5 and the soft fail clears, causing the event. If several points on the same card have soft fails during configuration, they cause the card to wait one minute before reporting the soft fail recoveries. Because of this delay, the event is not thrown away with other configuration events at step C above. The point has been functioning during the delay, and only the recovery message was delayed. These recovery events can be ignored.

If there are any I/O link failures during configuration or testing of an LLAI IOP, the IOP is removed from the list of LLAI IOPs to be tested and that action is reported.

TEST 1—During each pass, the Event Circular List is checked (see EVENT SCANNING) and all events are reported. If DEMAND_LOG is enabled, process values are read (from PV for TC points when FILTER_DELAY is greater than 0.0, and from PVALC in all other cases) for each point and displayed in a log in UNIT_OF_MEASURE terms. DEMAND_LOG then is disabled.

TEST 2—During each pass, the Event Circular List is checked (see EVENT SCANNING).

A.6.2 LLAI Alarm Codes

The final two digits of the alarm indicate the area where the failure was detected as follows:

- | | |
|-------|---|
| 00-30 | Failures while trying to build the IOP list used by the test. |
| 31-89 | Failures during configuration of the IOPs. |
| 90-99 | Failures while scanning the event list. |

A.7 STIM INTERACTIONS AND ALARM CODES

A.7.1 Hardware Interactions of STIM

INITIALIZATION—Each time the RUN command is entered, STIM performs the following before any tests are run:

- A. Checks the module type and status bytes (using READ SLOT PARAMETER) of each STIM IOP to be tested. If the module type is wrong or if the module status is bad, the reason(s) is reported and the Event Circular List is read and displayed, if possible. If the module type and status are both correct, the IOP is included in a list of IOPs to be tested.
- B. Checks for the availability of points at each IOP in the list created above. This is done by trying to upload from all points on the IOP (WRITE SLOT PARAMETER). When a configuration error is encountered, the point is not included in the points list.
- C. Configures each STIM IOP to be tested. This procedure consists of using WRITE SLOT PARAMETER to each slot to deactivate all slots, set deadband, frequency, EU span, span, and alarm trip points.
- D. Uses WRITE SLOT PARAMETER to place the card in the run mode.
- E. Sets each slot to active with a WRITE SLOT PARAMETER to PTEXECST.
- F. Discards any events in the circular event list.
- G. Gathers and reports any events, removes any IOPs that failed Configuration from the IOP test list and reports action taken.

TEST 1—During each pass the Event Circular List is checked (see EVENT SCANNING) and all events are reported. If DEMAND_LOG is enabled, a database is uploaded from each point and reported. DEMAND_LOG then is disabled. If LOAD XMTER is enabled, any changes required in HIVALUE and/or LOVALUE are downloaded to the database.

TEST 2—During each pass the Event Circular List is checked (see EVENT SCANNING) and all events are reported. If DEMAND LOG is enabled, a database is uploaded from each point and reported. DEMAND LOG then is disabled. Each pass causes random database changes.

TEST 3—Field utility test of any STIM device. This test is a menu-driven device communicator. The possible modifications to the database are URV, LRV, SFITAG, UPLOAD, DOWNLOAD, ACTIVATE, and DAMPING. Note that STIM must be 6-byte mode to run this test.

A.7.2 STIM Alarm Codes

The final two digits of the alarm indicate the area where the failure was detected as follows:

00-29	Failures while trying to build the IOP list used by the test.
31-32	Failures while trying to build the Point list used by the test.
33-49	Failures during configuration of IOPs.
50-59	Failures while trying to build the display log, typically a read slot parameter error.
60	Failure while checking Module Status.
65	Failure while scanning the Event List.
66	All points have been dropped from the Point list.
70-79	Failures while trying to effect changes in the database.
90-89	Failures while trying to Display Log.

A.8 PIMT INTERACTIONS AND ALARM CODES

A.8.1 Hardware Interactions of PIMT

NOTE

Because of memory constraints, DIMT and PIMT tests cannot run simultaneously.

INITIALIZATION—Each time the RUN command is entered, PIMT performs the following before any tests are run:

- A. Checks the module type and status bytes (using READ SLOT PARAMETER) of each PIM IOP to be tested. If the module type is wrong or if the module status is bad, the reason(s) is reported and the Event Circular List is read and displayed, if possible. If the module type and status are both correct, the IOP is included in a list of IOPs to be tested.
- B. Uses Send IOP Command to configure IOPs for 11-byte events.
- C. Configures each PIM IOP to be tested. This procedure uses WRITE SLOT PARAMETER to
 - 1. Set each IOP to Idle.
 - 2. Set each slot to "inactive" by writing to PTEXECST.
 - 3. Set each slot's trip points to NaN by writing to PVHHTP, PVHITP, PVLLTP, and PVLOTP.
 - 4. Set each slot's engineering and extended engineering unit ranges by writing to PVEXEUHI, PVEXEULO, PVEUHI, and PVEULO.
 - 5. Set each slot's scaling constants by writing to C1 and C2.
 - 6. Set each slot's timebase to seconds by writing to TIMEBASE.
 - 7. Set each slot's deadband EU value to zero by writing to PVALDBEU.
- D. Uses WRITE SLOT PARAMETER to place each IOP in the run mode.
- E. Sets each slot identified in POINT_MASK to "active" with a WRITE SLOT PARAMETER to PTEXECST.
- F. Discards any events in the circular event list from the PIM IOPs that were configured.
- G. Gathers and reports any events, removes any IOPs that failed configuration from the IOP test list, and reports action taken.

TEST 1—During each pass the Event Circular List is checked (see EVENT SCANNING at heading A.1.1) and all events are reported. If DEMAND_LOG is enabled, a report is generated depending on the value of REPORT_TYPE. DEMAND_LOG is disabled after a report is generated.

TEST 2—During each pass the Event Circular List is checked (see EVENT SCANNING at heading A.1.1) and all events are reported. If DEMAND_LOG is enabled, a report is generated. DEMAND_LOG is disabled after a report is generated.

A.8.2 PIMT Alarm Codes

TEST 01—REPORT

- 100-117 Failures while trying to build the IOP list used by the test.
- 120-144 Failures while trying to configure the IOPs.
- 150-153 Failures while scanning the Event List.
- 154 Failure while reading the initial counts.
- 160 Report rate alarm.
- 170-171 Report count alarms.

TEST 02—ACCURACY

- 200-217 Failures while trying to build the IOP list used by the test.
- 220-244 Failures while trying to configure the IOPs.
- 250-253 Failures while scanning the Event List.
- 260-273 Failures while setting the trip points on first pass.
- 274-276 Failures while checking the module status.
- 277-286 Failures while counting the number of out of tolerance alarms.

A.9 LLMX INTERACTIONS AND ALARM CODES

A.9.1 Hardware Interactions of LLMX

INITIALIZATION—Each time the RUN command is entered, LLMX performs the following before any tests are run:

- A. Checks the module type and status bytes (using READ SLOT COMMANDS) of each LLMX IOP to be tested. If the module type is wrong it is reported; or if the module status is bad, the reason(s) is reported (e.g., no FTA, soft failure, hard failure, etc.) and the Event Circular List is read in (with GET EVENT CIRCULAR LIST) and displayed, if possible. If the module type and status are both correct, the IOP is included in a temporary (local) list of LLMX IOPs to be tested. If either module type or status is wrong, the module is not placed in the temporary list.
- B. Configures each LLMX IOP to be tested. This procedure consists of using WRITE SLOT COMMANDS to each slot to deactivate all slots (parameter PTEXECST), set the engineering span (PVEULO, PVEUHI, PVEXEULO, PVEXEUHI), set the sensor type (SENSRTYP), set the characterization (PVCHAR), set the alarm deadband (PVALDB) to 0 percent, and set the required trip points (PVHITP, PVLOTP, PVHHTP, PVLLTP). If the current FTA type is 0-100 mVdc and UNITS_OF_MEASURE is TEMP, TF is set to the value specified in FILTER_DELAY and enables/disables Open Thermocouple Detection as specified in ENABLE/DISABLE; then, a WRITE SLOT PARAMETER to slot 0 (ModStat1) puts the module in RUN mode.
- C. All events in the Event Circular List are gathered and discarded because the events and changes created during configuration can be ignored.
- D. Reactivates the points by a WRITE SLOT PARAMETER to PTEXECST.
- E. Gathers and displays all Alarm Events in the event list.

If there are any I/O link failures during configuration or testing of an LLMX IOP, the IOP is removed from the temporary (local) list of LLMX IOPs to be tested and that action is reported.

TEST 1—During each pass the Event Circular List is checked (see EVENT SCANNING at heading A.1.1) and all events are reported (mask for all events is used). If DEMAND_LOG is enabled, process values are read (from PV for TC points when UNITS_OF_MEASURE is TEMP and FILTER_DELAY is not 0.0, and from PVCALC in all other cases) for each selected point on each FTA of the type(s) to be tested and displayed in a log in millivolts or degrees C for 0_100 mVdc/TC FTAs, depending on UNITS_OF_MEASURE, or degrees C for RTD FTAs. DEMAND_LOG then is disabled.

TEST 2—During each pass, the Event Circular List is checked (see EVENT SCANNING at heading A.1.1).

A.9.2 LLMX Alarm Codes

The final two digits of the alarm indicate the area where the failure was detected as follows:

- | | |
|-------|---|
| 00-30 | Failures while trying to build the IOP list used by the test. |
| 31-89 | Failures detected during configuration of the IOPs. |
| 90-99 | Failures detected while scanning the event list. |

A.10 IOPM INTERACTIONS AND ALARM CODES

A.10.1 Hardware Interactions of IOPM

INITIALIZATION—Each time the RUN command is entered, IOPM performs the following before any tests are run:

- A. Checks the status bytes (using READ SLOT PARAMETER) of each IOP to be tested. If the module status is bad, the reason(s) is reported and the Event Circular List is read and displayed, if possible. If the status is correct, the IOP is included in a list of IOPs to be tested.
- B. Sets IOPs to be tested to IDLE with a WRITE SLOT PARAMETER to slot 0.
- C. All events in the Event Circular List are gathered and discarded because the events and changes created during configuration can be ignored.
- D. Gathers and displays all Alarm Events in the event list.

If there are any I/O link failures during testing of an IOP, the IOP is removed from the list of IOPs to be tested and that action is reported.

TEST 1—During each pass the Event Circular List is checked (see EVENT SCANNING).

A.10.2 IOPM Alarm Codes

The final two digits of the alarm indicate the area where the failure was detected as follows:

- | | |
|-------|---|
| 00-69 | Failures while scanning the event list. |
| 70-99 | Failures while trying to build the IOP list used by the test, or failures during configuration of the IOPs. |

A.11 HIST INTERACTIONS AND ALARM CODES

A.11.1 Hardware Interactions of HIST

INITIALIZATION—Each time the RUN command is entered, HIST performs the following on the specified IOP before any testing is done:

- A. Checks the module type and status bytes (using READ SLOT PARAMETER commands) of the IOP. If the module type is wrong it is reported, or if either module status is bad, the reason(s) are reported (e.g., no FTA, soft failure, hard failure, etc.) and the Event Circular List is read in (with GET EVENT CIRCULAR LIST) and displayed, if possible. If module type and status are correct, testing continues. Otherwise, testing is aborted at this point.
- B. Next, the specified point(s) are configured. This consists of using WRITE SLOT PARAMETER commands to place the IOP in Idle mode (ModStat1), deactivate all points/slots (PTEXECST), set the engineering unit span (PVEULO, PVEUHI, PVEXEULO and PVEXEUHI), set the sensor type (SENSRTYP), set the characterization (PVCHAR), set the alarm deadband to 0 percent, set the required trip points (PVHITP, PVLOTP, PVHHTP, and PVLLTP), place the IOP in the RUN mode (ModStat1), and, finally, activate the specified points (PTEXECST).
- C. All the events in the Event Circular List are discarded because the events and changes created during configuration can be ignored.
- D. The appropriate number of histograms are reserved and initialized according to the settings of POINT_MASK, DISPLAY_OPT, and HISTOGRAM_OPT.

TEST 1—During each pass, the Event Circular List is checked (see EVENT SCANNING at heading A.1.1) and all events are reported. If necessary, waits for a scan time interval to have elapsed since the previous reading. For each specified point: reads PVRAW, evaluates and assimilates the value into the appropriate histograms. Any I/O link errors or invalid data (NaN) associated with a particular point results in an alarm and that point is dropped from further testing. Histogram reports are displayed according to the enable/disable settings of DEMAND_LOG and DISPLAY_AT_END_OF_RUN and the value chosen for PERIODIC_REPORT_PERIOD. INIT_HISTOGRAMS enable/disable determines whether or not the histograms are initialized after each report.

A.11.2 HIST Alarm Codes

The last two digits of the alarm indicate the area where the failure was detected as follows:

00-30	Failures during initialization while trying to validate the IOP type and status.
31-89	Failures during initialization while trying to configure the points on the IOP.
90-99	Failures detected while scanning the event list or reading the point values.

A.12 DSOE INTERACTIONS AND ALARM CODES

A.12.1 HARDWARE INTERACTIONS OF DSOE

INITIALIZATION—Each time the RUN command is entered, DISOE performs the following before any tests are run:

- A. Checks the module type and status bytes (using READ SLOT PARAMETER) of each DISOE IOP to be tested. If the module type is wrong or if the module status is bad, the reason is reported and the ECL is read and displayed, if possible. If the module type and status are both correct, the IOP is included in a list of IOPs to be tested.
- B. Configures each DISOE IOP to be tested. This procedure uses READ SLOT PARAMETER and WRITE SLOT PARAMETER to:
 1. Set each DISOE IOP to IDLE.
 2. Set each DISOE IOP's points to INACTIVE.
 3. Set each DISOE IOP point's parameters to their corresponding Unique Parameter value.
 4. Assign a logical address to each DISOE IOP.
 5. Set each DISOE IOP to RUN.
 6. Clear each DISOE IOP's PVCL by writing the append pointer value to the remove pointer.

TEST FIRST PASS—Each test performs the following “standard” configuration (using READ SLOT PARAMETER and WRITE SLOT PARAMETER), in addition to any test specific setup, on each DISOE IOP to be tested:

- A. Set the appropriate DISOE IOP points to ACTIVE.
- B. Clear each DISOE IOP's ECL by writing the append pointer value to the remove pointer.

TEST 1—Once each pass the PVCL is read.

TEST 2—Once each pass the PVCL is read.

TEST 3—Once each pass the PVCL is read.

PASS COMPLETION FOR EACH PASS—After each test has completed a pass, the DISOE IOP's ECL is read.

A.12.2 DSOE ALARM CODES

DSOE MAIN

001 - 019	Failures while trying to build the IOP test list.
020 - 028	Failures while trying to write the IOP slot parameters.
029 - 032	Failures while trying to read and report ECL events.
033	Failure while performing one pass of the tests in the test list.

Test 1

101 - 102	Failures while performing First Pass test setup.
103	Failure while reporting extraneous event.
104 - 106	Failures while collecting the PVCL events.
107	Failure while analyzing the PVCL events.

Test 2

201 - 202	Failures while performing First Pass test setup.
203 - 205	Failures while collecting the PVCL events.
206	Failure while processing the PVCL events.

Test 3

301 - 302	Failures while performing First Pass test setup.
303 - 305	Failures while collecting the PVCL events.
306	Failure while counting the PVCL events.

A.13 SIMT ALARMS

SIMT alarms are explained in subsection 13.7.

FAX Transmittal

FAX No.: (602) 313-4212

TO: W. Hukow

Total FAX pages: _____
(including this page)

Reader Comments

Title of Document: Process Manager Test System

Document Number: PM13-610

Issue Date: 07/00

Comments: _____

Recommendations: _____

FROM:

Name: _____ **Date:** _____

Title: _____

Company: _____

Address: _____

City: _____ **State:** _____ **ZIP:** _____

Telephone: _____ **FAX:** _____

FOR ADDITIONAL ASSISTANCE:

Write	Call
Honeywell Inc. Industrial Automation and Control 16404 North Black Canyon Highway Phoenix, AZ 85053	Technical Assistance Center (TAC) 1-800-822-7673 (48 contiguous states except Arizona) 602-313-5558 (Arizona)

